Communication and Cooperation with Human and Artificial Agents

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ABSTRACT

People delegate decisions to artificial intelligence (AI), especially large language models, which act and communicate on their behalf. Many such interactions are social dilemmas, in which individually rational choices leave everyone worse off than if all had cooperated. Studies find that allowing communication among participants in social dilemmas increases the number of people working for the mutual benefit of the group. In this study, we test how people cooperate in one-shot Prisoner's Dilemma and Stag Hunt games when their partner is an AI agent playing for another person. We compare conditions featuring AI agents with and without communication to control conditions featuring human-to-human interactions, both with and without communication. Participants attributed far more agency to human than to AI partners and trusted them slightly more. Yet communication mattered far more for cooperation than partner type, as it raised cooperation by 19 to 31 percentage points with either partner, while partner type had no significant effect within any condition. Communication detectably raised trust only with human partners. This suggests that people bring an instrumental rather than a relational theory of mind to machines: they cooperate when an AI partner signals that it will, without trusting it as they would a person.

1. Introduction

Social dilemmas—situations in which individually rational choices produce a collectively suboptimal outcome (Kollock, 1998)—have been a central object of study across psychology, economics, political science, and biology. Communication among participants in social dilemmas promotes cooperation and other-regarding behavior, leaving the group better off (Balliet, 2010; Orbell, Van de Kragt and Dawes, 1988; Ostrom, Walker and Gardner, 1992). Why communication has this effect is less settled. One long-standing account is that communication builds mutual *trust* (Orbell et al., 1988; Ostrom et al., 1992). Trust involves a willingness to accept vulnerability based on a positive assessment of the partner's intentions (Mayer, Davis and Schoorman, 1995; Yamagishi and Yamagishi, 1994), and communication provides evidence of those intentions. A second, complementary account is *belief-based*: many people are *conditional cooperators*—individuals who cooperate if and only if they expect their partner to do the same (Fischbacher, Gächter and Fehr, 2001; Fischbacher and Gächter, 2010)—and pre-play communication, in a controlled social dilemma experiment, raises that expectation by signaling cooperative intent. In ordinary human interaction these channels are tied together and difficult to tell apart. A human partner in a cooperation scenario that seems trustworthy will also seem likely to cooperate.

As agentic artificial intelligence rises and becomes embedded in every aspect of social life, the nature of cooperation in social dilemmas is bound to change. People increasingly make decisions alongside, and through, artificial agents, and they do not treat a machine as they treat another person. Research suggests that humans reason about the trustworthiness of an artificial agent differently than about trusting a human (Hancock, Billings, Schaefer, Chen, De Visser and

Parasuraman, 2011; Schniter, Shields and Sznycer, 2020; Stanton and Jensen, 2021; Glikson and Woolley, 2020; Fahnenstich, Rieger and Roesler, 2024), and they attribute a different kind of mind to it (Gray, Gray and Wegner, 2007; Epley, Waytz and Cacioppo, 2007; Halevy, 2016; Camerer, Ho and Chong, 2015). Normally, people tend to spontaneously infer the beliefs, intentions, and goals of social partners—a capacity known as *theory of mind* (Premack and Woodruff, 1978). This capacity is central to human cooperation, but it operates differently, and less reliably, when the partner is known to be an artificial system. These differences are important due to their behavioral consequences: a review of 162 studies finds that people tend to act more selfishly toward artificial agents (March, 2021), and more broadly people respond systematically differently to algorithmic than to human judgment, sometimes avoiding it and sometimes over-relying on it (Dietvorst, Simmons and Massey, 2015; Logg, Minson and Moore, 2019).

This raises the central question of our study: when a person can communicate with an AI agent before choosing in a social dilemma, does communication promote cooperation as it does between humans, and if so, through which channel—by building interpersonal trust, or by shifting beliefs about the agent's likely cooperation? The distinction matters because the kind of mind people grant a machine may support one channel but not the other. The question is both newly pressing and newly testable. It is pressing because LLM-mediated communication is poised to affect social trust, cooperation, and the foundations of political and economic order, as autonomous agents increasingly act and communicate on behalf of the people they represent. It is testable because large language models (LLMs), unlike the scripted agents of earlier work, can exchange natural-language messages and so communicate in a social dilemma much as a human partner would.

Researchers have studied strategic interactions between humans and artificial agents since at least the 1970s, but the literature leaves this question open. Most such studies use simple, deterministic algorithms; very few use artificial intelligence (AI) or LLMs (e.g., Akata, Schulz, Coda-Forno, Oh, Bethge and Schulz, 2025; Erlei, Nekdem, Meub, Anber and Waldschmidt, 2020; Jakesch, French, Ma, Hancock and Naaman, 2019). Ng (2023) called for comparisons with and without communication and for measures of expected cooperation. We address both by varying communication across human and AI partner conditions and eliciting beliefs about the partner's choice.

2. Experimental Design

We conducted two studies. A pilot study, which is described in more detail below, and the main study. We describe the main study first because it is the focus of our exposition and discussion. We also discuss the differences between the two studies in detail, though they are broadly similar.

In the main study we feature standard, one-shot Prisoner's Dilemma and Stag Hunt games[1] embedded in a between-subjects 2x2 factorial design. One factor is the type of partner a participant is matched with: a human agent, or an AI agent playing on behalf of a passive human agent. The second factor manipulates the presence or absence of an initial communication stage, in which players can choose from a set of pre-determined messages to send to their counterparts. Participants played both the Prisoner's Dilemma and Stag Hunt game variants, in that order, after being initially randomly sorted into one of four conditions:

- **Human–No Comm**: Human–human interaction, without communication
- **Human–Comm**: Human–human interaction, with communication
- **AI–No Comm**: Human–AI agent interaction, without communication
- **AI–Comm**: Human–AI agent interaction, with communication

The design is pre-registered at OSF Registries.[2] The study underwent IRB review and received approval (see Appendix N).

There is no meaningful way to pay an AI agent. However, withholding payments from another human would introduce one more difference from the human-human condition, to the extent that prosocial motivations drive cooperation (Schniter et al., 2020). Indeed, a common rationale for employing artificial agents is precisely to remove social preferences from the decision (Houser and Kurzban, 2002; Johnson, Camerer, Sen and Rymon, 2002). Instead, we deliberately preserve them by paying a passive human beneficiary. We explicitly and truthfully informed participants in the hybrid human–AI conditions that they were playing against an AI agent that was making decisions on behalf of the other player, and that the other player had no control over the agent's decisions but would be paid based on them. We did this to isolate the effect of the player's type. In each condition, we recruited between 89 and 127 participants from MTurk Prime and Connect (CloudResearch) (see Appendix Table C.2 for the precise sample sizes). The participation payment was $1.50 for MTurk Prime participants and $2.00 for Connect participants, to which we added any bonus that the players earned. We also paid the passive humans on whose behalf the AIs played, according to the decisions the AIs made.

The payoff tables below detail the aforementioned 2x2 design and were presented to participants to make clear the bonuses they could get by playing the game.[3] Payoffs are consistent with the accepted forms for both the Prisoner's

[1] In the Prisoner's Dilemma, defecting is each player's best response to either choice the partner can make, yet mutual defection leaves both players worse off than mutual cooperation: individually rational play undermines the collective interest. In the Stag Hunt, there is no single best response: cooperating (Stag) pays most if the partner also cooperates, while the safe option (Hare) is the better choice if the partner does not. The Stag Hunt is thus technically a coordination game rather than a social dilemma, but the operative variable is the same: an expectation about the partner's cooperative intent.

[2] Link to OSF Registration: [OSF link withheld for blind review]

[3] In both games, Choice A corresponds to defection and Choice B to cooperation. The experiment avoided loaded terms throughout: participants saw only the neutral labels Choice A and Choice B, and the games were introduced simply as Game 1 and Game 2, never by their names. Note that,

Table 1
Payoff table for Prisoner's Dilemma games (all conditions)

		The Other Participant	
		Choice A	Choice B
You	**Choice A**	$1.25, $1.25	$2.50, $0.50
	Choice B	$0.50, $2.50	$1.75, $1.75

Table 2
Payoff table for Stag Hunt games (all conditions)

		The Other Participant	
		Choice A	Choice B
You	**Choice A**	$1.25, $1.25	$1.75, $0.50
	Choice B	$0.50, $1.75	$2.50, $2.50

Dilemma and Stag Hunt games. In the Prisoner's Dilemma conditions, Choice A represents defection and Choice B represents cooperation. In the Stag Hunt conditions, Choice A represents Hare and Choice B represents Stag.

For participants paired with an AI agent, we wanted to show the capabilities of ChatGPT to play strategy games without inducing priming bias that could happen if we showed an example of ChatGPT playing each of the two games in real time. Thus, we showed the participants screenshots and a GIF of us prompting the ChatGPT to play tic-tac-toe. These files can be found in the attached appendix.

LLMs, like humans, can generate essentially unlimited amounts of text of varying length and content. That is part of what makes them so powerful. However, allowing either type of game partner to generate messages freely would introduce uncontrolled variation in message content. Any difference in behavior throughout conditions could then reflect either the partner manipulation or differences in the messages participants happened to receive. Instead, we provided participants with the following seven messages to choose from before deciding between A or B:

In the Prisoner's Dilemma conditions.

1. Let's work together to maximize our rewards. If we both choose Option B, we can earn $1.75 each. This way, neither of us will be left out, and we'll both gain equally. Shall we agree on that?
2. We have an opportunity to choose a path that benefits both of us. If we both select Option B, we'll both earn $1.75 each. This seems like the fairest approach, don't you think?
3. Choosing Option A might be tempting, but it could also mean that one of us ends up with very little. If we both commit to Option B, we'll both earn a fair reward. Let's choose collaboration over competition.
4. Our choices can reflect our values. Let's make a pact to choose Option B, ensuring that we both benefit equally. We can make the fair and ethical choice, can't we?

for example, in the Prisoner's Dilemma, if both players choose B they each earn $1.75, yet if one defects to A the defector earns $2.50 and the other player only $0.50; if both choose A, each earns $1.25.

5. I will choose Option A.
6. I will choose Option B.
7. I prefer to send no message.

In Stag Hunt conditions.

1. I am an individual that values trust and loyalty, and I want to believe you also value those principles; let's show that by choosing B together.
2. I care about your success just as much as mine, and working together helps us both win. Choosing B lets us both score a good bonus.
3. Let's make sure neither of us gets the short end of the stick; teaming up is the fair way to go. Let's both choose B.
4. Imagine the frustration of missing out because one of us didn't coordinate on the best choice; let's avoid that by both choosing B.
5. I will choose Option A.
6. I will choose Option B.
7. I prefer to send no message.

The first four messages in all conditions were created by prompting OpenAI's GPT-3.5 Turbo model to think about arguments favoring cooperation. The LLM was first asked to define a complete structure for each game and define a utility function based on both quantitative and qualitative factors, in separate threads for the Prisoner's Dilemma and the Stag Hunt. Then the LLM created messages appealing to identity, altruism, fairness, cognitive dissonance aversion, and social norms. Note that, due to the stochastic nature of ChatGPT's responses, using the same prompt multiple times results in similar but not identical responses. More detail on the prompting process can be found in the attached appendix. To the set of LLM-generated messages, we added the two simple options "I will choose Choice A" and "I will choose Choice B," as well as an option to send no message.

The study was programmed using Python via the oTree behavioral experiment framework. Before entering the online main study, participants read the consent information on screen and clicked "I agree to participate in this study" to continue. No signature was requested, consistent with the IRB's waiver of signed documentation of consent. In all conditions, participants saw instructional templates detailing the game's rules and progression before any actions. We also asked a simple question regarding payoffs given an arbitrary pair of decisions to ensure comprehension of the rules. We showed a new set of instructions before the second (Stag Hunt) game and emphasized that the payoffs of the game had changed, to ensure that participants played both games correctly. In conditions including pre-play communication, neither participant could continue with the game until they had both sent and received a message. Message selection

was done simultaneously, and only when both players selected a message did a follow-up screen reveal both messages. Then we asked participants to select either Choice A or Choice B, with a reminder of which messages were sent and received. In non-communication conditions, participants were simply asked to select between Choice A and Choice B. In both conditions, participants were told what their total payoff would be only after both games had been played, with a reminder of which decisions had resulted in that payoff.

2.1. Additional Survey Items

We also included additional survey items after participants had finished the game sequence. These included game-specific measures of beliefs about partner cooperation, trustworthiness, and agency, as well as standard demographic and political orientation questions and items measuring participants' familiarity with, use of, and attitudes toward artificial intelligence. The full text of all survey questions is available in Appendix B.

2.2. AI Agent Decision Making

We utilized OpenAI's GPT-3.5 Turbo model to simulate the behavior of an LLM player. The model was prompted with a dynamically constructed string object that provided the game scenario and instructions, mirroring the information presented to human participants. This ensured that the LLM, for the purposes of the experiment, was treated as if it were a human player capable of understanding text and making choices.

Our approach to prompting was twofold. First, we allowed for randomness in the LLM's responses to emulate its behavior in the widely used ChatGPT online platform. To achieve this, we set up the model temperature equal to 1, and we repeatedly probed the LLM with the same prompt and recorded its textual responses. Second, to improve the LLM's reasoning, we appended a sentence to the prompt that encouraged step-by-step and chain-of-thought reasoning. This was based on literature suggesting that such prompts improve LLMs' problem-solving abilities (e.g., Renze and Guven, 2024). We instructed the model to enclose its answer within brackets, to simplify the extraction of the LLM's choices from its responses. This allowed us to systematically process the response text and code them as choices using regular expressions.

We made over three thousand API queries with the OpenAI Python library, to model the relative likelihood of Choice A and Choice B in our AI agent's decision-making. This process provided us with the LLM's distribution of choices for each game condition and for every possible combination of messages sent in communication conditions. Our findings revealed that the LLM's behavior was consistent with human participants: without communication in the Prisoner's Dilemma, the LLM chose to cooperate 46% of the time. However, the LLM's cooperation rate surged to 90% when a communication phase was introduced. This behavior is consistent with existing research that suggests communication, especially messages appealing to fairness, can boost cooperation rates (e.g., Balliet, 2010; Bicchieri and Lev-On, 2007; Bohnet and Frey, 1999).

Participants assigned to an AI partner condition were explicitly told that they were playing against an AI agent acting on behalf of another person. Before playing, they saw a demonstration of ChatGPT playing tic-tac-toe to illustrate how we elicited its responses to game instructions. During synchronous play, the AI agent sampled its choices from probabilities estimated from GPT-3.5 Turbo for the corresponding game and condition. Where communication was available, it also sampled a message and conditioned its decision on the messages sent and received. Using these saved probabilities held the response policy fixed across sessions and avoided the latency and runtime dependencies of a live API connection.

2.3. Relationship to Pilot Study

The main study was preceded by a pre-registered pilot study (OSF `[ID withheld]`, $N = 447$) conducted via Qualtrics using the strategy method (Brandts and Charness, 2011), in which participants made contingent cooperation decisions for each of seven possible messages. The pilot study employed a Prisoner's Dilemma only, used an asynchronous survey design, and included two additional delegator conditions.[4] Three pilot findings shaped the present study's design. First, communication increased cooperation regardless of partner type. Second, the communication boost was more than twice as large for AI partners (+38.6 pp) as for human partners (+17.4 pp)—the opposite of the pre-registered prediction that communication would *decrease* cooperation when a participant is paired with an AI agent. Third, partner type had a significant effect in no-communication conditions (23.6% cooperation with AI partners vs. 46.6% with human partners, $p = .004$), which raised the question of whether this baseline aversion would persist under synchronous, incentivized interaction. These results motivated the shift to oTree's real-time protocol, the addition of the Stag Hunt, and the revised hypothesis that communication would help AI conditions *more*, not less. Full pilot study methods, analysis, and results are reported in Appendix D.

3. Hypotheses

The following hypotheses are based on those pre-registered alongside the experimental design.[5]

3.1. Cooperation

Prior studies point in two directions on cooperation with artificial partners. A review of 162 experimental studies finds that people often act more selfishly toward computer players (March, 2021). They also cooperate less with a bot that discloses its nature than with one that passes as human (Ishowo-Oloko, Bonnefon, Soroye, Crandall, Rahwan and Rahwan, 2019). Yet humans apply social rules and expectations to computers as well (Nass and Moon, 2000), and H1

[4] By asynchronous, we mean that participants completed the survey individually, without a partner deciding at the same time. Decisions were nonetheless incentivized. After data collection, we randomly paired each participant's choice with a partner's choice and paid bonuses according to the resulting payoffs.

[5] Some hypothesis language has been lightly edited for clarity. Deviations from the registration are documented in Appendix M. See Appendix E for the full formal notation and inequalities.

follows the larger of the two bodies of evidence. Communication raises cooperation in social dilemmas (Balliet, 2010), and it did so with a computer partner in an early Prisoner's Dilemma experiment (Kiesler, Sproull and Waters, 1996). H2 expects the same here. H3 adds that a pre-play message has more room to raise cooperation with an AI partner if distrust depresses the baseline there.

H1 — AI partner reduces cooperation. Cooperation rates will be lower when a participant's counterpart is an AI agent compared to the corresponding human–human condition. Tested separately for each game-by-communication combination (H1a–d).

H2 — Communication increases cooperation. In both human and AI conditions, the presence of a pre-play communication stage will increase cooperation rates relative to the corresponding no-communication condition. Tested separately for each game-by-partner-type combination (H2a–d).

H3 — Communication effect is larger with an AI partner. The boost in cooperation from adding communication will be greater in human–AI conditions than in human–human conditions (H3a–b).

3.2. Trust

People trust machines on different grounds than they trust other people (Glikson and Woolley, 2020), and their emotional responses to a robot partner differ even when their trust investments do not (Schniter et al., 2020), which grounds H4. Trust depends in part on a partner's perceived benevolence (Mayer et al., 1995), and H5 expects communication to raise it with either partner by signaling good intentions. However, readers judge a host as less trustworthy when they believe AI wrote the host's profile and see it among profiles written by people (Jakesch et al., 2019). H6 therefore predicts that the trust gap persists under communication.

H4 — Lower trust with AI partner. Pooling across communication conditions, participants will report lower trust in an AI agent than in a human partner.

H5 — Communication increases trust. Within each partner type, the communication condition will yield higher trust ratings than the no-communication condition (H5a–b).

H6 — Trust gap persists under communication. Even with communication, trust will be higher in the human–human condition than in the human–AI condition.

3.3. Agency and Regulation

In the classic study of mind perception, adults placed a robot far below adult humans on agency (Gray et al., 2007). H7 tests that gap in a strategic setting. For H8, we expected that interacting with an AI counterpart, and the risk of being exploited by it, would make the stakes of delegating decisions to machines salient. A recent field experiment,

however, found that direct experience with an AI boss changed job performance but not attitudes toward AI in public decisions (Margalit and Raviv, 2026).

H7 — Higher agency attribution to humans. Participants will attribute greater agency (intentions, having a mind of one's own) to human counterparts than to AI agents, comparing the two partner types within each communication condition (H7a–b).

H8 — AI exposure increases regulation support. Participants in AI conditions will express stronger support for government regulation of AI than participants in the corresponding human–human conditions (H8a–b).

4. Results

A total of 460 participants completed the experiment and were distributed across four conditions: Human–No Comm ($n = 127$), Human–Comm ($n = 127$), AI–No Comm ($n = 117$), and AI–Comm ($n = 89$). Below, we report results organized by dependent variable: cooperation rates in the two social dilemmas, post-game trust, perceived agency of the other player, and support for AI regulation. All pairwise comparisons use the Tukey–Kramer procedure (Kramer, 1956), which generalizes Tukey HSD for unequal group sizes (see Appendix F for details); effect sizes are reported as Cohen's d for pairwise contrasts and partial eta-squared (η_p^2) for omnibus tests.

Note on the pilot study. The pilot study's full results are reported in the Appendix, including mixed-effects models, marginal effects, and strategy-method cooperation profiles. The main pilot findings—communication as the dominant factor, the reversal of the AI communication prediction, and the elimination of the partner-type gap under communication—all replicate in the present data. We focus below on main study results; cross-study comparisons appear in the Discussion.

4.1. Cooperation

Prisoner's Dilemma. Communication dramatically increased cooperation in the Prisoner's Dilemma, whereas partner type had no significant effect in either communication condition. A one-way ANOVA across the four conditions was significant, $F(3, 456) = 16.76$, $p < .001$, $\eta_p^2 = .099$. Cooperation in the Prisoner's Dilemma averages roughly 70% across the two communication conditions against roughly 41% across the two no-communication conditions, as shown in the left panel of Figure 1.

Contrary to H1, partner type did not reduce cooperation. Cooperation in the Human–No Comm condition ($M = 44.9\%$) did not differ significantly from AI–No Comm ($M = 36.8\%$; $p = .539$, $d = .17$), and cooperation in Human–Comm ($M = 75.6\%$) did not differ from AI–Comm ($M = 64.0\%$; $p = .294$, $d = .25$). Neither H1a nor H1b is supported.[6]

[6]Pooled across communication conditions, the mixed-effects model estimates 10.2 percentage points lower Prisoner's Dilemma cooperation with AI partners ($p = .023$; Table A.5). The pre-registration specified only the per-condition tests reported here.

Supporting H2, communication significantly increased cooperation for both partner types. Among human-partner dyads, cooperation rose from 44.9% without communication to 75.6% with it ($p < .001$, $d = .66$), supporting H2a. Among AI-partner dyads, cooperation rose from 36.8% to 64.0% ($p < .001$, $d = .56$), supporting H2b. The absolute communication boost was slightly larger with human partners (30.7 percentage points) than with AI partners (27.3 percentage points), contrary to the direction predicted by H3a; the difference between these two boosts was not statistically significant (see Table A.5).

Stag Hunt. The Stag Hunt showed a broadly similar pattern, with communication again serving as the primary driver of cooperation. A one-way ANOVA was significant, $F(3, 456) = 12.55$, $p < .001$, $\eta_p^2 = .076$. The right panel of Figure 1 displays the condition means: cooperation was highest in AI–Comm (89.9%) and lowest in AI–No Comm (61.5%), with the two human-partner conditions falling in between (Human–No Comm = 67.7%; Human–Comm = 86.6%).

Partner type did not significantly affect cooperation, as in the Prisoner's Dilemma. Human–No Comm (67.7%) did not differ from AI–No Comm (61.5%; $p = .650$, $d = .13$), and Human–Comm (86.6%) did not differ from AI–Comm (89.9%; $p = .941$, $d = .10$). H1c and H1d are not supported.

Communication increased cooperation for both partner types. Communication raised cooperation with human partners by 18.9 percentage points ($p = .002$, $d = .46$), supporting H2c. For AI-partner dyads, communication produced a boost of 28.4 percentage points, from 61.5% to 89.9% ($p < .001$, $d = .68$), supporting H2d.

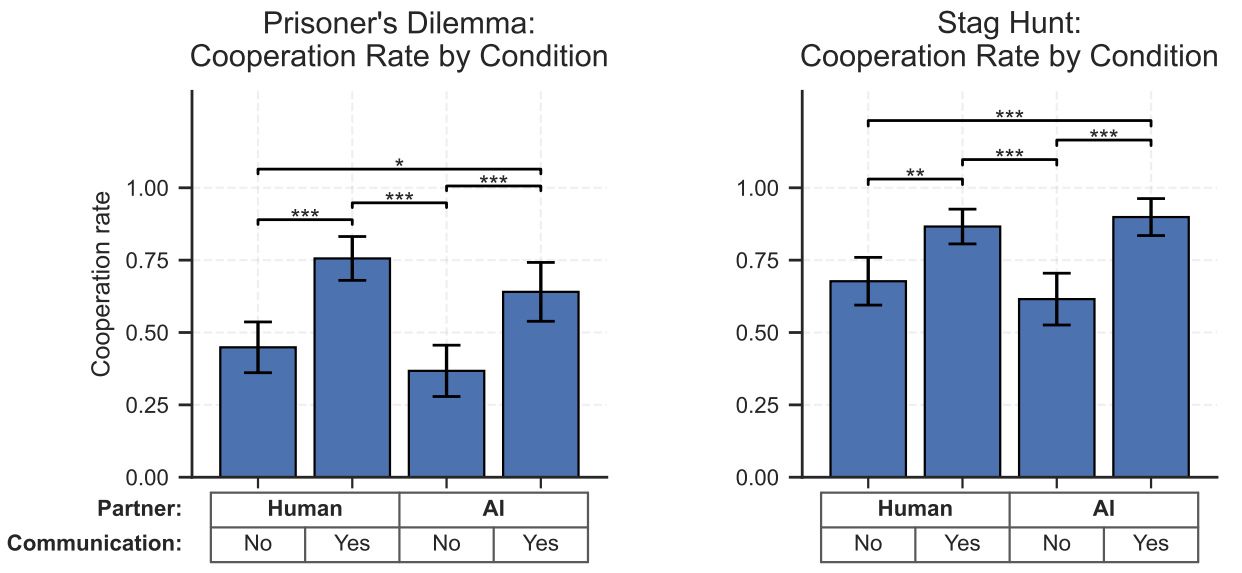


Figure 1: Cooperation rate by condition in the Prisoner's Dilemma (left) and Stag Hunt (right). In both games, communication substantially increases cooperation; partner type (human vs. AI) has no significant effect. Error bars show 95% CIs; brackets indicate significant pairwise differences (Tukey–Kramer). Significance markers: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

Factorial analysis. We estimated a mixed-effects linear probability model with a random intercept and random slope for game type per participant, plus a centered participant-level control for passing the PD comprehension check, to decompose the 2 (Partner Type) × 2 (Communication) × 2 (Game) design while accounting for the within-subject dependence across games (full model in Table A.4). The model shows the patterns visible in Figure 2. Communication was the dominant predictor ($b = .306$, $z = 5.15$, $p < .001$). Stag Hunt cooperation was higher than Prisoner's Dilemma cooperation overall ($b = .228$, $z = 5.20$, $p < .001$), consistent with the payoff-dominant equilibrium favoring cooperation.[7] Partner type had no significant effect in the no-communication Prisoner's Dilemma, the model's reference condition ($b = -.084$, $z = -1.38$, $p = .168$), and the Partner Type × Communication interaction was not significant in either the Prisoner's Dilemma ($b = -.035$, $z = -0.40$, $p = .691$) or the Stag Hunt ($b = .093$, $z = 1.19$, $p = .233$).

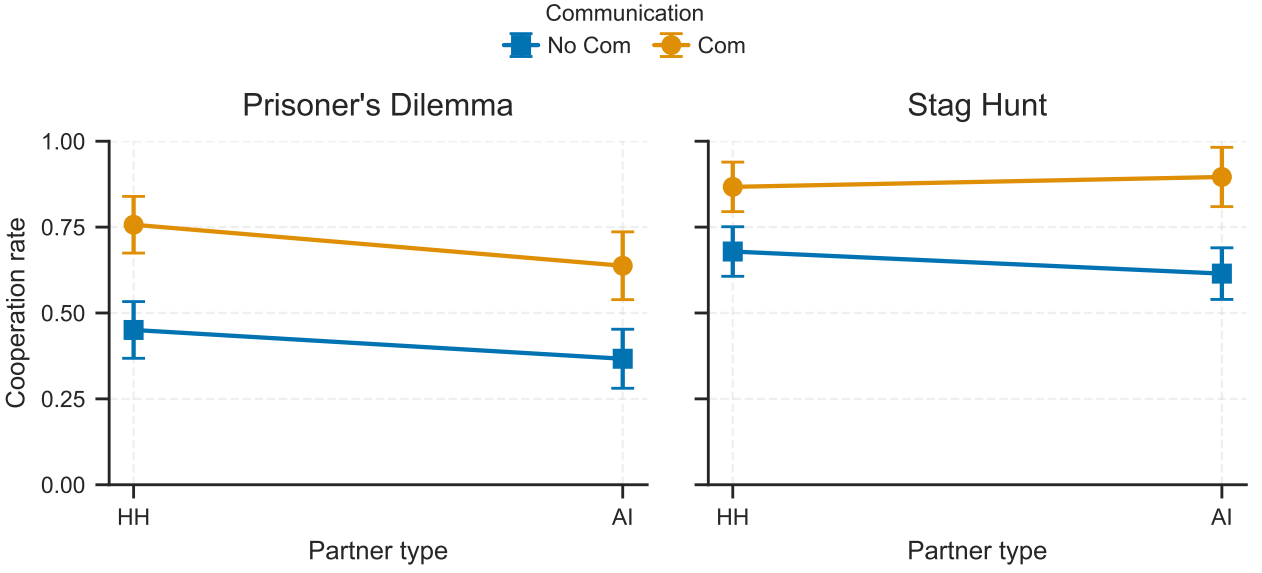


Figure 2: Cooperation rate by partner type and communication. Communication substantially increases cooperation in both games; the effect of partner type (human vs. AI) is small and non-significant. Error bars show 95% CIs.

In summary, the cooperation results clearly support H2 (communication increases cooperation) but do not support H1 (AI partner reduces cooperation). H3 predicted that communication helps more in AI conditions. The interaction held in the predicted direction in the Stag Hunt only. In the Prisoner's Dilemma, the boost was slightly larger with human partners. Neither interaction reached conventional levels of statistical significance in the mixed-effects model reported above (Table A.4).

4.2. Trust

Participants reported their trust in the other player on a single post-experiment Likert item (1 = *Very Untrustworthy*, 5 = *Very Trustworthy*). Participants paired with an AI partner reported lower trust than those paired with a human

[7] An equilibrium is payoff dominant when it yields every player a higher payoff than any other equilibrium (Harsanyi and Selten, 1988); in the Stag Hunt, that is mutual cooperation (Stag).

partner, consistent with H4. Pooling across communication conditions, the human-partner mean was 3.47, compared with 3.29 for those paired with an AI partner ($t(427.42) = 2.43$, $p = .015$, $d = .23$). The effect is modest in magnitude and statistically significant. H4 is supported.

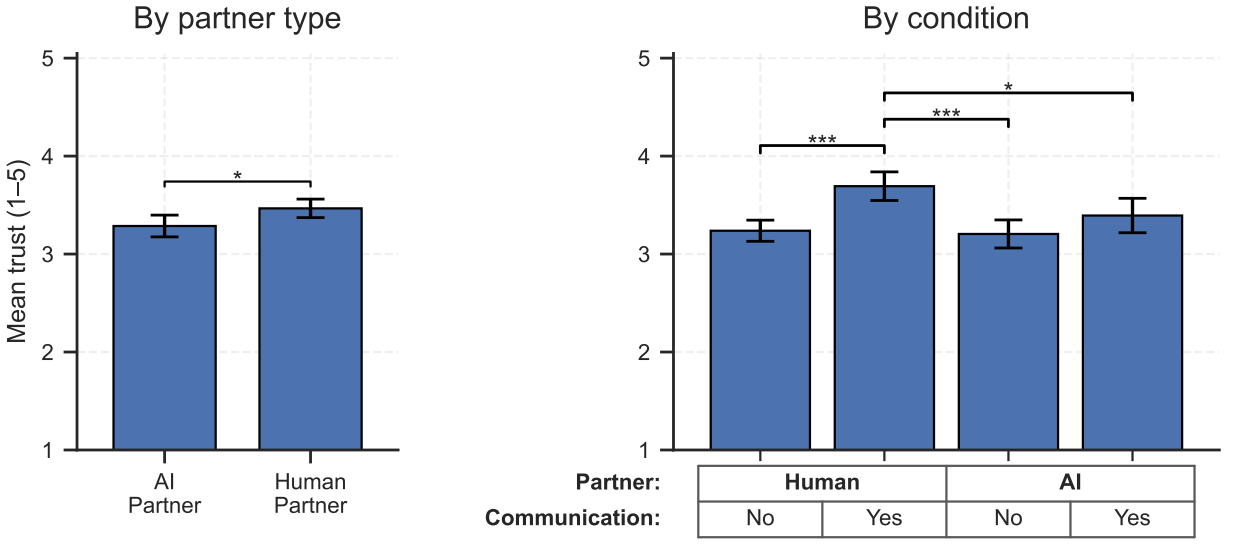


Figure 3: Trust in the other player (single post-experiment item). Left: AI partners are trusted less than human partners ($p = .015$, $d = .23$), consistent with the expectation. Right: communication significantly increases trust in the human-partner conditions but not in the AI-partner conditions; this asymmetry suggests different mechanisms of trust-building. Error bars show 95% CIs; brackets indicate significant pairwise differences (Tukey–Kramer). Significance markers: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

A one-way ANOVA across the four conditions was significant, $F(3, 455) = 10.57$, $p < .001$, $\eta_p^2 = .065$, indicating that trust varied meaningfully when conditions were examined individually. The pattern reflects an asymmetric communication effect, as the right panel of Figure 3 illustrates. Communication significantly increased trust in human-partner conditions: Human–Comm ($M = 3.69$) was rated substantially higher than Human–No Comm ($M = 3.24$; $p < .001$, $d = .62$), supporting H5a. In contrast, communication did not significantly increase trust in AI-partner conditions: AI–Comm ($M = 3.39$) was not significantly higher than AI–No Comm ($M = 3.21$; $p = .300$, $d = .23$). H5b is not supported.

H6 predicted that even with communication, trust in human partners would exceed trust in AI partners. Trust in the Human–Comm condition ($M = 3.69$) was significantly higher than in AI–Comm ($M = 3.39$; $p = .025$, $d = .36$). H6 is therefore supported. The Tukey comparisons also reveal an even larger contrast between Human–Comm and AI–No Comm ($p < .001$, $d = .60$), indicating that the combination of a human partner and communication produced the highest trust levels in the study.

Taken together, the trust results reveal something counterintuitive: communication boosted trust with human partners but not with AI partners, even as it boosted *cooperation* with both. This dissociation suggests that the

mechanism through which communication promotes cooperative behavior with AI partners may not operate through increased interpersonal trust.

4.3. Perceived Agency

Participants rated the perceived agency of their partner using a two-item scale averaging "has intentions" and "has a mind of its/their own" (1–5 Likert, 1 = *Strongly Disagree*, 5 = *Strongly Agree*; Cronbach's $\alpha = .86$; see Appendix B for the full item set). Participants attributed more agency to human than to AI partners, and this gap was the largest effect observed in this study. As shown in Figure 4, participants attributed substantially more agency to human partners ($M = 3.99$) than to AI partners ($M = 2.43$; $t(382.18) = 14.80$, $p < .001$, $d = 1.42$), strongly supporting H7. This is a large effect by conventional benchmarks: the two distributions are separated by more than one standard deviation.

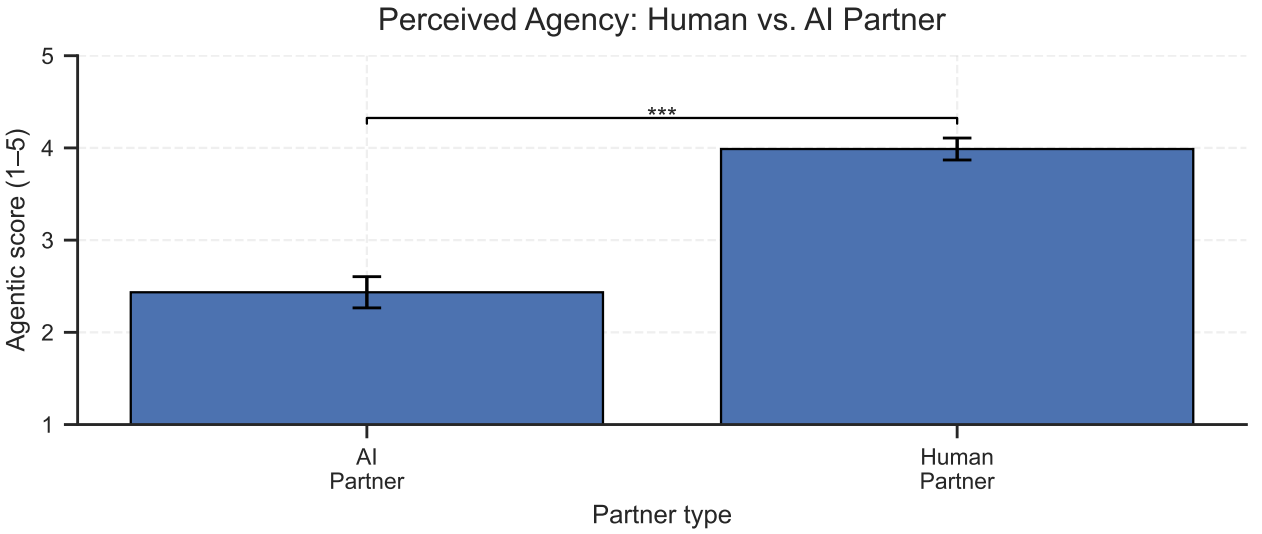


Figure 4: Perceived agency of the other player by partner type. Participants attribute substantially more agency to human partners than to AI partners ($p < .001$, $d = 1.42$), the largest effect in the study. Error bars show 95% CIs; the bracket marks the partner-type difference. Significance markers: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

A four-condition ANOVA shows that the effect is driven entirely by partner type rather than communication, $F(3, 455) = 77.26$, $p < .001$, $\eta_p^2 = .337$. Within human-partner conditions, agency scores were nearly identical (Human–No Comm $M = 3.98$ vs. Human–Comm $M = 3.99$; $p = 1.000$, $d = .01$), and the same was true within AI-partner conditions (AI–No Comm $M = 2.35$ vs. AI–Comm $M = 2.55$; $p = .543$, $d = .17$). Every cross-type comparison was highly significant (all $p < .001$, all $d \geq 1.29$). Participants thus clearly distinguished between human and AI partners on the agency dimension, regardless of whether communication was present.

The large gap in perceived agency did not translate into a corresponding gap in cooperation or trust. Participants attributed far less mind and intentionality to AI partners, yet cooperated with them at similar rates and reported comparable levels of trust—a pattern we return to in the Discussion.

4.4. Conditional Cooperation Beliefs

After each game, participants rated their perceived likelihood that the other player (or the other player's AI agent) had chosen B—that is, cooperated (1 = *Extremely Unlikely*, 5 = *Extremely Likely*). These belief items capture whether participants expected their partner to reciprocate, and they are correlated with the cooperation results.

In the Prisoner's Dilemma, a one-way ANOVA revealed significant differences in beliefs across conditions, $F(3, 455) = 9.08$, $p < .001$, $\eta_p^2 = .056$. The pattern mirrors the cooperation findings: participants in communication conditions expected more cooperation from their partners. The Human–Comm condition had the highest beliefs ($M = 3.89$), significantly above both no-communication conditions (vs. Human–No Comm: $p < .001$, $d = .62$; vs. AI–No Comm: $p = .002$, $d = .47$). The AI–Comm condition ($M = 3.69$) was intermediate.

In the Stag Hunt, the same pattern held, $F(3, 455) = 10.24$, $p < .001$, $\eta_p^2 = .063$. Communication conditions again yielded higher cooperation beliefs. The Human–Comm ($M = 4.16$) and AI–Comm ($M = 4.11$) conditions clustered above AI–No Comm ($M = 3.61$; Human–Comm vs. AI–No Comm: $p < .001$, $d = .61$; AI–Comm vs. AI–No Comm: $p < .001$, $d = .60$). Partner type produced no significant differences in beliefs—participants expected similar levels of cooperation from human and AI partners within the same communication condition.

These belief results are consistent with the behavioral data: communication raised both actual cooperation and beliefs about partner cooperation, while partner type affected neither (Figure 5).

We also dichotomized each rating as a "high" belief—a 4 or 5, that the partner was *likely* or *very likely* to have cooperated—to place beliefs on the same scale as the cooperation decision. We then modeled the resulting share with the same mixed-effects specification used for cooperation (PD and SH items stacked, with a random intercept and slope per participant). Appendix Figure A.1 plots the predicted shares on the same 0–1 scale and in the same format as the cooperation interaction plot (Figure 2). They follow the behavioral results closely (full coefficients in Appendix Table A.8). Communication raised the share of high beliefs in both games—from .52 to .72 in the Prisoner's Dilemma and from .63 to .81 in the Stag Hunt with human partners—but in the Prisoner's Dilemma this increase was smaller for AI partners (.53 to .67); in the Stag Hunt the communication boost was equally large for both partner types. This dichotomization is a presentational device, adopted so that beliefs can be read on the same 0–1 scale as the cooperation decision; the inferential test of the interaction is carried by the model on the underlying 1–5 ratings, reported in the Discussion and in Appendix Table A.9.

4.5. Support for AI Regulation

Our final measure moves from how people behave with an AI partner to what they want done about AI in general. We asked whether direct experience with an AI agent shifts support for regulating AI (1 = *No Regulation*, 7 = *Full Regulation*), linking the individual-level interaction studied here to the macro-level governance debate.

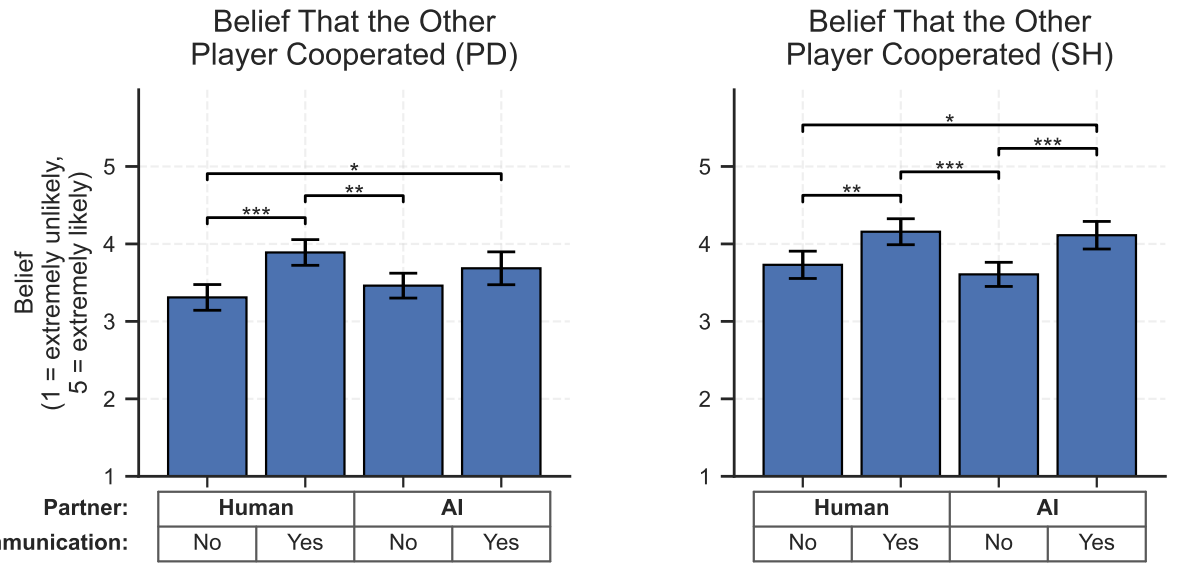


Figure 5: Beliefs about partner cooperation by condition (post-game item: "How likely do you think the other player chose B?"). The pattern mirrors the behavioral cooperation results: communication raises beliefs, partner type does not. Error bars show 95% CIs; brackets indicate significant pairwise differences (Tukey–Kramer). Significance markers: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

H8 predicted that participants in the AI-partner conditions would express stronger support for government regulation of AI than participants in the human-partner conditions. This hypothesis was not supported. A one-way ANOVA across the four conditions yielded $F(3, 456) = 2.67$, $p = .047$, $\eta_p^2 = .017$. As Figure 6 shows, mean regulation support was similar across conditions: Human–No Comm ($M = 4.68$), Human–Comm ($M = 5.12$), AI–No Comm ($M = 4.87$), and AI–Comm ($M = 5.15$). No pairwise comparison reached conventional significance (smallest $p = .079$, all $|d| < .32$). H8a and H8b are not supported.

No individual pairwise contrast reached the conventional threshold, although the omnibus ANOVA was significant ($F(3, 456) = 2.67$, $p = .047$). AI-partner conditions were numerically more supportive of regulation than human-partner conditions, in line with the predicted direction, but the difference did not reach statistical significance. Brief strategic interaction thus appears insufficient to shift the stable political and ideological orientations that likely underlie attitudes toward AI governance.

Summary. Across both social dilemmas, communication was the dominant driver of cooperation, raising rates by 19–31 percentage points depending on the game and partner type. Partner type (human vs. AI) had no significant effect on cooperation or cooperation beliefs within any condition, a small effect on trust ($d = .23$), and the study's largest effect on perceived agency ($d = 1.42$). Communication increased both cooperation and beliefs about partner cooperation regardless of partner type, but increased trust only with human partners—pointing to different psychological pathways.

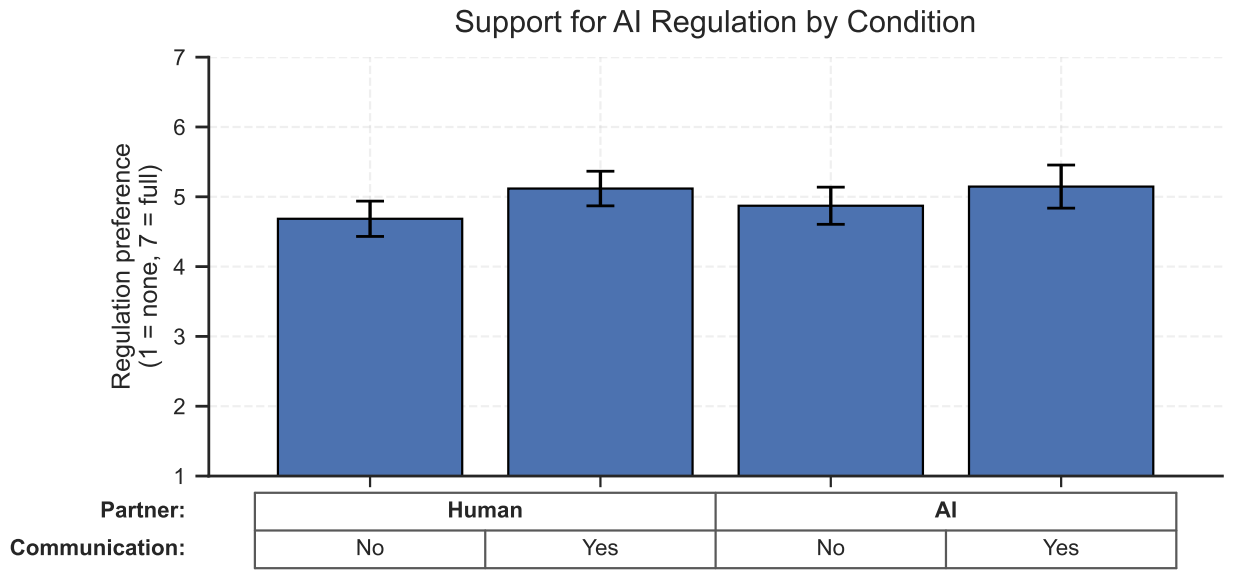


Figure 6: Support for AI regulation by condition. Scale: 1 = no regulation, 7 = full regulation. Error bars show 95% CIs.

Support for AI regulation showed a small omnibus shift across conditions ($p = .047$) with no individual pairwise contrast reaching the conventional threshold.

5. Discussion

Our findings speak to how humans interact with AI agents in social dilemmas, and to the role of communication in promoting cooperation. We found no significant differences in cooperation rates between human-human and human-AI conditions in either the Prisoner's Dilemma or Stag Hunt games (H1a-d), contrary to our initial hypotheses and previous literature suggesting that people are more selfish when playing against artificial agents (March, 2021). This unexpected result suggests that when AI agents are acting on behalf of passive human beneficiaries, participants may maintain similar levels of prosocial behavior as they would with direct human counterparts.

Communication promoted cooperation, strongly supporting H2. Both human–human and human–AI conditions showed significantly higher cooperation rates with communication in the Prisoner's Dilemma (H2a–b), and the same held for both partner types in the Stag Hunt (H2c–d). The communication boost was directionally larger with AI partners than with human partners in the Stag Hunt, but slightly larger with human partners in the Prisoner's Dilemma; in neither game did the Partner Type × Communication interaction reach significance (PD: $p = .691$; SH: $p = .233$; Table A.5), so H3a and H3b were not supported. Where the direction holds (Stag Hunt), it is consistent with the possibility that structured communication closes whatever cooperation deficit AI partners might otherwise produce.

Trust dynamics revealed additional complexity in human-AI interactions. As predicted by H4, participants reported slightly lower trust in AI partners than in human partners ($d = .23$, $p = .015$)—a modest but statistically significant

difference. The impact of communication on trust, however, varied between human and AI conditions: communication significantly increased trust in human-human interactions (H5a) but had no significant effect on trust in human-AI interactions (H5b). This asymmetry suggests that while communication can increase cooperative behavior with AI agents, it may operate through different psychological mechanisms than in human-human interactions.

The agency results reveal a surprising disconnection between perceived agency and other seemingly related beliefs and behaviors. Participants clearly distinguished human from AI partners on the agentic scale (H7; $d = 1.42$, $p < .001$, the largest effect in the study), yet this stark perceptual gap did not translate into equally large differences in cooperation, trust, or cooperation beliefs. Communication did not further shift agency attributions within either partner type ($p > .54$). Support for AI regulation did not move in the way H8 predicted: participants paired with an AI partner were no more supportive of regulating AI than those paired with a human (H8a–b not supported). The omnibus ANOVA was marginally significant ($F(3, 456) = 2.67$, $p = .047$), but no pairwise contrast reached the conventional threshold; the comparison with the smallest adjusted p-value—itself short of significance—was between the two human conditions (No Comm vs. Comm, $p = .080$, $d = .30$), a communication effect orthogonal to the AI-exposure prediction. Brief strategic interaction thus appears not to shift stable political and ideological orientations toward AI in a targeted way.

Cross-study convergence. The pilot study and the main study together strengthen confidence in the robustness of these findings. The pilot used the strategy method with the PD only, and the main study used direct response with both the PD and the Stag Hunt. Both studies identify communication as the dominant driver of cooperation, with effect sizes ranging from 17 to 39 percentage points across conditions, methods, and games. Both studies disconfirm the prediction that AI partners would uniformly depress cooperation. And both show that once communication is introduced, the AI cooperation deficit vanishes.

The one instructive divergence concerns the partner-type effect without communication. The pilot study detected a significant baseline aversion to AI partners in no-communication conditions ($b = -0.23$, $p = .004$), which was absent in the main study ($b = -.084$, $p = .168$). One explanation is methodological: the strategy method's "cold" deliberative framing may amplify sensitivity to partner identity, whereas direct response under real-time social presence compresses it. An alternative is that synchronous interaction (knowing the other player is deciding *right now*) reduces the psychological distance between human and AI partners. Either way, the pattern reinforces the paper's central conclusion: under ecologically valid conditions, communication matters and partner type does not.

A conditional-cooperation account, and what it implies for theory of mind about machines. The clearest theoretical reading of our results is that communication raised cooperation with AI partners through a *belief* channel rather than a *trust* channel. Three results must be considered together. First, communication raised cooperation with AI partners to a level statistically indistinguishable from that observed with human partners. Second, communication

raised participants' beliefs that their partner would cooperate, and it did so for AI partners as much as for human partners in the Stag Hunt—where AI–Comm and Human–Comm conditions were statistically indistinguishable, both well above AI–No Comm—though in the Prisoner's Dilemma the belief boost was significantly smaller for AI partners (partner × communication interaction $p = .042$; Appendix Table A.7). Third, communication raised *trust* only for human partners; with AI partners trust did not move (H5b). Cooperation and beliefs rose together while reported trust stayed flat.

This is what a conditional-cooperation model predicts (Fischbacher et al., 2001; Fischbacher and Gächter, 2010): many people cooperate to the extent that they expect their counterpart to cooperate. A pre-play message that signals cooperative intent updates that expectation directly; it need not generate interpersonal trust to change behavior, only make the partner's cooperative move expected. For human partners the message appears to do both, raising beliefs and trust together; for AI partners it does the first without the second. Communication with an AI agent thus seems to operate through conditional cooperation that never surfaces as reported trust.

This dissociation maps onto the structure of the mind people attribute to machines. Mind perception is organized along two dimensions—*agency* (the capacity to plan, intend, and act) and *experience* (the capacity to feel) (Gray et al., 2007; Epley et al., 2007). Our participants drew an extremely sharp line between human and AI partners on the agency-laden items (intentions, having a mind of one's own; $d = 1.42$, the largest effect in the study), yet this gap depressed neither beliefs nor cooperation. We read this as people granting an AI counterpart enough agency to be treated as a predictable strategic actor whose signaled intentions are worth conditioning on, but not the experiential, relational mind that interpersonal trust is built on (for a similar asymmetry in children, see Xu, Thomas, Yu and Pan, 2025). Reported trust depends on the experience dimension people do not grant to machines. Conditional cooperation depends on the agency dimension they do grant. On this account, humans bring an instrumental rather than relational theory of mind to cooperation with machines: predict-and-respond rather than relate-and-trust.

That AI partners were cooperated with at all—at rates indistinguishable from humans, despite the agency gap—is itself consistent with the long-standing finding that people apply social rules and expectations to computers even while explicitly denying that the machine has a human-like mind (Nass and Moon, 2000). What our design adds is the mechanism: the social response survives because it runs through belief-based conditional cooperation, a channel that does not require the machine to be trusted in the way a person is. A practical corollary is that designing AI systems to communicate structured cooperative intentions may be more productive than attempting to replicate the trust-building dynamics of human interaction.

Implications for AI deployment in coordination settings. These results are relevant to the growing deployment of autonomous AI agents that act and communicate on behalf of humans in strategic settings (Chen, Guan, Elshafiey,

Zhao, Zekeri, Shaibu, Prince and Wu, 2026), a capability now available through general-purpose agent frameworks that route an AI system's decisions and messages on a user's behalf with limited real-time human oversight. In our experimental design, an AI agent makes game decisions and exchanges pre-play messages on behalf of a passive human beneficiary, which mirrors this arrangement. Our results suggest that the design of systems in which AI agents interact with humans or with other agents on behalf of humans should prioritize the quality and structure of communication protocols over whether a counterpart is human or artificial.

Limitations and future directions. Several design choices constrain generalizability. First, participants selected from a fixed menu of seven pre-written messages rather than composing free-form text; this controlled for message content but prevents conclusions about natural-language communication with AI agents. Second, both games were one-shot; repeated interaction may activate reputation and reciprocity mechanisms that shift the relative importance of partner type and communication. Third, participants are U.S. online workers recruited via MTurk Prime and Connect, and cross-cultural variation in trust and AI attitudes may moderate these effects. Fourth, participants knew they were in an experiment; field studies embedding AI coordination partners in natural task environments would strengthen external validity. A productive next step would be a study that crosses free-form vs. structured communication with human vs. AI partners in a repeated game, allowing researchers to test whether the trust-cooperation dissociation we observe persists when participants can build (or fail to build) a communication history with their AI counterpart over multiple rounds.

Data availability

The data and analysis code that support the findings of this study are deposited in Harvard Dataverse and will be made publicly available upon publication. The repository link is withheld here to preserve anonymity during review: the deposit names the authors, so citing it would defeat blind review. The deposit is currently private and its DOI does not yet resolve.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used Claude Code (Anthropic), running the Claude Opus 5 and Claude Fable 5 models, in order to generate code snippets for data cleaning and statistical analysis, which the authors thoroughly inspected and modified before use and validated for correctness and appropriateness. The authors designed the original experiment and selected the statistical models, estimation methods, software packages, and tests used in the analysis. The authors wrote the first complete version of the manuscript without AI assistance. Claude Code (version

2.1.283; Anthropic), running the Claude Opus 5.5 model, was then used to generate excerpts and revise sections of the manuscript. The final text combines passages written by the authors, AI-generated excerpts, and AI-generated passages subsequently edited by the authors. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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Appendix

Communication and Cooperation with Human and Artificial Agents

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A. Statistical Tables

This appendix reports the full statistical tables underlying the results in the main text, grouped by study and, within each study, by analysis.

A.1. Pilot Study — Strategy-Method Design (Qualtrics)

The pilot used the strategy method: participants stated a contingent cooperation decision for each possible message rather than interacting in real time. The tables below report its cooperation rates, model coefficients, and pre-registered marginal effects.

A.1.1. Predicted Cooperation Rates

Table A.1 reports the cooperation rate in each of the four conditions, with 95% confidence intervals and the number of participants per cell.

A.1.2. Mixed-Effects Model Coefficients

Table A.2 regresses cooperation on partner type and communication in a mixed-effects linear probability model with a participant random intercept.

A.1.3. Marginal Effects (Pre-Registered Hypotheses)

Table A.3 translates that model into the contrasts named in the pre-registration: the effect of communication within each partner type, and the effect of partner type within each communication condition.

A.2. Main Study — Synchronous Design (oTree)

The main study replaced the strategy method with real-time incentivized play and added the Stag Hunt, so each participant contributes one decision per game. The tables below follow the same order as the pilot, then add the factorial, belief, covariate-adjusted, and logistic specifications.

A.2.1. Mixed-Effects Model Coefficients

Table A.4 reports the main cooperation model: the $2 \times 2 \times 2$ treatment structure with a random intercept and a random slope for game type per participant. This is the model whose coefficients are quoted in the Results.

A.2.2. Mixed-Model Marginal Effects

Table A.5 reports the marginal effects implied by that model—the communication and partner-type contrasts within each game, on the percentage-point scale used in the main text.

Table A.1
Cooperation rates by condition with 95% confidence intervals (pilot study).

	No Communication	Communication
Human–Human	46.6% [34.9, 58.3]	63.9% [57.0, 70.8]
Human–AI	23.6% [13.6, 33.7]	62.2% [54.0, 70.3]

Mean [95% CI]. N per cell in parentheses where applicable.

Table A.2
Mixed-effects linear probability model: cooperation regressed on partner type and communication (pilot study).

	Dep. var.: Cooperation (binary)				
	Coef.	**SE**	z	p	**95% CI**
Baseline (reference group)	0.4658	0.0563	8.28	<.001***	[0.355, 0.576]
Partner type (1 = AI, 0 = Human)	-0.2296	0.0798	-2.88	0.004**	[-0.386, -0.073]
Communication (1 = yes, 0 = no)	0.1735	0.0670	2.59	0.010**	[0.042, 0.305]
Partner type × Communication	0.2120	0.0956	2.22	0.027*	[0.025, 0.399]
Observations	1223				
Participants	299				
σ_u^2 (random intercept)	0.0854				
σ_ε^2 (residual)	0.1456				
ICC	0.370				

$^{***}p < .001$ $^{**}p < .01$ $^{*}p < .05$ $^{\dagger}p < .10$

A.2.3. Factorial Regressions (Non-Cooperation DVs)

The following table reports OLS regressions of each non-cooperation dependent variable on the 2×2 factorial structure (partner type × communication). For cooperation beliefs, a separate mixed-effects model stacking PD and SH beliefs with participant random intercepts is also reported.

A.2.4. Cooperation Beliefs: Interaction Plot

Figure A.1 presents the stacked-belief mixed-effects model graphically, in the same interaction format as the cooperation plot (Figure 2). It shows the predicted share of participants who rated the other player *likely* or *very likely* (a rating of 4 or 5) to have chosen B. The figure conveys the same information as the condition-level beliefs in Figure 5 and is reported here to avoid duplication in the main text.

Tables A.8 and A.9 report the full mixed-effects coefficients behind this figure: the dichotomized high-belief share, and the underlying raw 1–5 rating, respectively, both on the same specification used for cooperation (PD and SH stacked, with a random intercept and slope per participant, and a centered control for PD comprehension).

A.2.5. Covariate-Adjusted Models

As a robustness check and to address the pre-registered plan for "logistic regression with demographic covariates," we re-estimated the main cooperation model with seven pre-treatment covariates: age, gender (female indicator),

Table A.3
Effect of partner type and communication on cooperation: marginal effects from a mixed-effects model with participant random intercepts (pilot study, strategy method).

Partner type effect (H1)							
	Human–Human	**Human–AI**	**Δpp**	**SE**	z	p	**95% CI**
AIH cooperation < HH (partner type effect, pooled)[a]	55.3%	42.9%	-12.4	4.8	-2.59	0.010**	[-21.7, -3.0]
Partner type effect (no-comm only)[b]	46.6%	23.6%	-23.0	8.0	-2.88	0.004**	[-38.6, -7.3]
Partner type effect (comm only)[c]	63.9%	62.2%	-1.8	5.3	-0.34	0.737	[-12.1, +8.5]
Communication effect (H2, H3)							
	No Communication	**Communication**	**Δpp**	**SE**	z	p	**95% CI**
Communication effect (HH)[d]	46.6%	63.9%	+17.4	6.7	2.59	0.010**	[+4.2, +30.5]
Communication effect (AIH)[e]	23.6%	62.2%	+38.5	6.8	5.66	<.001***	[+25.2, +51.9]
Interaction							
	Comm effect (HH)	**Comm effect (AIH)**	**Δpp**	**SE**	z	p	**95% CI**
Comm effect AIH − Comm effect HH[f]	17.4%	38.6%	+21.2	9.6	2.22	0.027*	[+2.5, +39.9]

$^{***}p < .001$ $^{**}p < .01$ $^{*}p < .05$ $^{\dagger}p < .10$
Δpp = percentage-point change in P(cooperate). CIs are Wald (±1.96× SE).
[a] OSF: Δ < 0 (AIH cooperates less than HH) — CONFIRMED
[b] OSF: Δ < 0 (AIH cooperates less than HH) — CONFIRMED
[c] OSF: Δ < 0 (AIH cooperates less than HH) — NOT CONFIRMED (correct direction, not significant)
[d] OSF: Δ > 0 (communication increases HH cooperation) — CONFIRMED
[e] OSF: Δ < 0 (communication DECREASES AIH cooperation) — OPPOSITE DIRECTION (significant)
[f] OSF: Δ < 0 (comm effect smaller/negative for AIH) — OPPOSITE DIRECTION (significant)

Table A.4
Mixed-effects linear probability model: cooperation regressed on partner type, communication, and game type with participant random intercepts and slopes (main study).

	Dep. var.: Cooperation (binary)				
	Coef.	**SE**	z	p	**95% CI**
Baseline (reference group)	0.4506	0.0421	10.70	<.001***	[0.368, 0.533]
Partner type (1 = AI, 0 = Human)	-0.0838	0.0608	-1.38	0.168	[-0.203, 0.035]
Communication (1 = yes, 0 = no)	0.3063	0.0595	5.15	<.001***	[0.190, 0.423]
Game type (1 = Stag Hunt, 0 = Prisoner's Dilemma)	0.2283	0.0439	5.20	<.001***	[0.142, 0.314]
Partner type × Communication	-0.0355	0.0894	-0.40	0.691	[-0.211, 0.140]
Partner type × Game type	0.0195	0.0634	0.31	0.758	[-0.105, 0.144]
Communication × Game type	-0.1181	0.0621	-1.90	0.057†	[-0.240, 0.004]
Partner type × Comm × Game (three-way)	0.1287	0.0933	1.38	0.168	[-0.054, 0.311]
Comprehension check passed (centered control)	-0.0508	0.0525	-0.97	0.333	[-0.154, 0.052]
Observations	920				
Participants	460				
σ^2_u (random intercept)	0.1287				
σ^2_ε (residual)	0.0961				
ICC	0.573				

$^{***}p < .001$ $^{**}p < .01$ $^{*}p < .05$ $^{\dagger}p < .10$

education (college degree or higher), political ideology (1–7 scale), AI awareness (1–5), AI usage (mean of three items, 1–5), and log-transformed completion time. These covariates were selected because they are measured independently of the experimental manipulation and could plausibly predict cooperation.

Table A.5
Effect of partner type, communication, and game type on cooperation: marginal effects from a mixed-effects model with participant random intercepts and slopes (main study).

Partner type effect							
	Human–Human	**Human–AI**	**Δpp**	**SE**	z	p	**95% CI**
Partner type effect (PD, no comm)[a]	45.1%	36.7%	-8.4	6.1	-1.38	0.168	[-20.3, +3.5]
Partner type effect (PD, comm)[b]	75.7%	63.7%	-11.9	6.6	-1.82	0.069†	[-24.8, +0.9]
Partner type effect (SH, no comm)[c]	67.9%	61.5%	-6.4	5.3	-1.21	0.226	[-16.9, +4.0]
Partner type effect (SH, comm)[d]	86.7%	89.6%	+2.9	5.7	0.50	0.616	[-8.4, +14.1]
Partner type effect (PD, pooled across comm)[e]	60.4%	50.2%	-10.2	4.5	-2.27	0.023*	[-18.9, -1.4]
Partner type effect (SH, pooled across comm)[e]	77.3%	75.5%	-1.8	3.9	-0.45	0.650	[-9.5, +5.9]
Communication effect							
	No Comm	**Comm**	**Δpp**	**SE**	z	p	**95% CI**
Communication effect (PD, HH)[f]	45.1%	75.7%	+30.6	6.0	5.15	<.001***	[+19.0, +42.3]
Communication effect (PD, AIH)[g]	36.7%	63.7%	+27.1	6.7	4.06	<.001***	[+14.0, +40.1]
Communication effect (SH, HH)[h]	67.9%	86.7%	+18.8	5.2	3.62	<.001***	[+8.6, +29.0]
Communication effect (SH, AIH)[i]	61.5%	89.6%	+28.1	5.8	4.82	<.001***	[+16.7, +39.6]
Game effect							
	Prisoner's Dilemma	**Stag Hunt**	**Δpp**	**SE**	z	p	**95% CI**
Game effect (SH − PD, pooled)[j]	55.3%	76.4%	+21.1	2.3	9.06	<.001***	[+16.6, +25.7]
Interaction: Communication × Partner type							
	Comm effect (HH)	**Comm effect (AIH)**	**Δpp**	**SE**	z	p	**95% CI**
comm × partner (PD)[k]	30.6%	27.1%	-3.5	8.9	-0.40	0.691	[-21.1, +14.0]
comm × partner (SH)[l]	18.8%	28.1%	+9.3	7.8	1.19	0.233	[-6.0, +24.6]

$^{***}p < .001$ $^{**}p < .01$ $^{*}p < .05$ $^{\dagger}p < .10$
Δpp = percentage-point change in P(cooperate). CIs are Wald (±1.96× SE).
[a] H1a — OSF: $\Delta < 0$ (AIH cooperates less) — NOT CONFIRMED (correct direction, not significant)
[b] H1b — OSF: $\Delta < 0$ (AIH cooperates less) — NOT CONFIRMED (correct direction, not significant)
[c] H1c — OSF: $\Delta < 0$ (AIH cooperates less) — NOT CONFIRMED (correct direction, not significant)
[d] H1d — OSF: $\Delta < 0$ (AIH cooperates less) — NOT CONFIRMED (opposite direction)
[e] Pooled across communication conditions; not a pre-registered test.
[f] H2a — OSF: $\Delta > 0$ (comm increases cooperation) — CONFIRMED
[g] H2b — OSF: $\Delta > 0$ (comm increases cooperation) — CONFIRMED
[h] H2c — OSF: $\Delta > 0$ (comm increases cooperation) — CONFIRMED
[i] H2d — OSF: $\Delta > 0$ (comm increases cooperation) — CONFIRMED
[j] OSF: $\Delta > 0$ (higher cooperation in SH) — CONFIRMED
[k] H3a — OSF: $\Delta > 0$ (comm effect larger for AIH in PD) — NOT CONFIRMED (opposite direction)
[l] H3b — OSF: $\Delta > 0$ (comm effect larger for AIH in SH) — NOT CONFIRMED (correct direction, not significant)
Hypothesis numbers follow the formal specifications in Appendix E.

The covariate-adjusted estimates are nearly identical to the base model. The treatment coefficients change by less than 0.02 in absolute value, and the residual variance reduction is minimal (0.7%), indicating that the covariates explain very little of the remaining variance once treatment and random effects are accounted for. This is consistent with successful randomization: the experimental conditions are balanced on observables, and the treatment effects are not confounded.

A.2.6. Logistic Specification Check

Cooperation is binary, and the models above treat it linearly (see Appendix F for why). To confirm that this choice does not drive any conclusion, we re-estimated the covariate-adjusted cooperation model as a logistic regression with standard errors clustered on the participant. Clustering plays the role the random intercept plays in the linear model:

Table A.6
OLS regressions of non-cooperation dependent variables on the 2×2 factorial structure (partner type × communication; main study).

	(1) *Trust*	(2) *Agency*	(3) *Honest/Fair*	(4) *Belief (PD)*	(5) *Belief (SH)*	(6) *Regulation*
Baseline (reference group)	3.238***	3.984***	3.095***	3.309***	3.730***	4.685***
	(0.068)	(0.097)	(0.061)	(0.084)	(0.082)	(0.128)
Partner type (1 = AI, 0 = Human)	-0.033	-1.638***	0.050	0.152	-0.123	0.187
	(0.098)	(0.140)	(0.088)	(0.121)	(0.118)	(0.185)
Communication (1 = yes, 0 = no)	0.455***	0.008	0.373***	0.580***	0.427***	0.433*
	(0.096)	(0.137)	(0.086)	(0.118)	(0.116)	(0.181)
Partner type × Communication	-0.267†	0.196	-0.367**	-0.356*	0.078	-0.159
	(0.144)	(0.206)	(0.129)	(0.177)	(0.174)	(0.272)
Observations	459	459	459	459	459	460
R^2	0.065	0.338	0.049	0.057	0.063	0.017

$^{***}p < .001$ $^{**}p < .01$ $^{*}p < .05$ $^{\dagger}p < .10$. Standard errors in parentheses.

Table A.7
Mixed-effects model: cooperation beliefs (PD + SH stacked) on partner type, communication, and game type (main study).

	Dep. var.: Cooperation belief, mixed-effects (1|pid)				
	Coef.	**SE**	z	p	**95% CI**
Baseline (reference group)	3.3095	0.0829	39.92	<.001***	[3.147, 3.472]
Partner type (1 = AI, 0 = Human)	0.1520	0.1195	1.27	0.203	[-0.082, 0.386]
Communication (1 = yes, 0 = no)	0.5802	0.1170	4.96	<.001***	[0.351, 0.810]
Partner type × Communication	-0.3564	0.1756	-2.03	0.042*	[-0.701, -0.012]
Game type (1 = Stag Hunt, 0 = Prisoner's Dilemma)	0.4206	0.0844	4.99	<.001***	[0.255, 0.586]
Partner type × Game type	-0.2753	0.1216	-2.27	0.024*	[-0.514, -0.037]
Communication × Game type	-0.1529	0.1191	-1.28	0.199	[-0.386, 0.080]
Partner type × Comm × Game (three-way)	0.4346	0.1786	2.43	0.015*	[0.085, 0.785]
Observations	918				
Participants	459				
σ^2_u (random intercept)	0.4178				
σ^2_ε (residual)	0.4483				
ICC	0.482				

$^{***}p < .001$ $^{**}p < .01$ $^{*}p < .05$ $^{\dagger}p < .10$

it absorbs the dependence between a participant's two decisions, one in each game. The specification is otherwise identical—the same $2 \times 2 \times 2$ treatment structure, the same seven covariates, the same 920 observations from 460 participants.

Table A.13 reports the estimates. Every treatment term agrees with the linear model in *both* sign and significance. Communication raises cooperation ($b = 1.36$, $p < .001$) and the Stag Hunt elicits more cooperation than the Prisoner's Dilemma ($b = 0.99$, $p < .001$); the partner-type effect is negative but not significant ($b = -0.39$, $p = .169$), as is the partner × communication interaction ($b = -0.17$, $p = .679$). Because logit coefficients are log-odds, their magnitudes are not comparable to the linear model's percentage points, and only the pattern of signs and significance is informative. That pattern is unchanged, so nothing in the results rests on the linear specification.

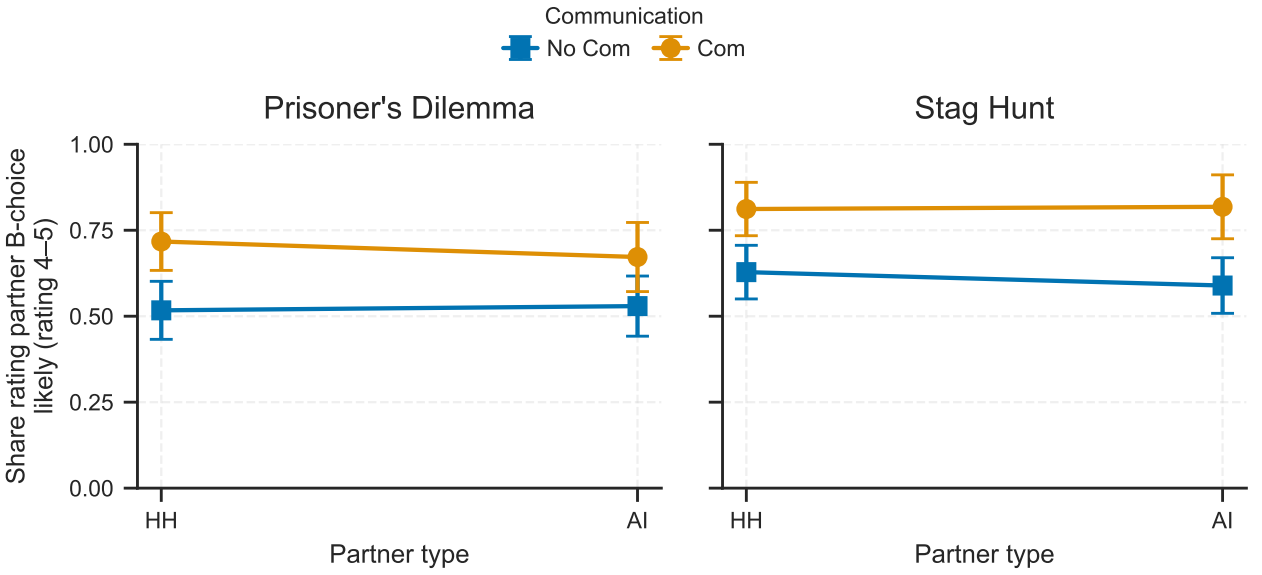


Figure A.1: Model-predicted share of participants rating the other player *likely* or *very likely* to have chosen B (cooperated)—a belief rating of 4 or 5—from a mixed-effects model with the PD and SH belief items stacked and a random intercept and slope per participant. Plotted on the same 0–1 scale and in the same format as the cooperation interaction plot (Figure 2): partner type on the x-axis, communication conditions as separate lines. Communication raises expected cooperation for both partner types; in the Prisoner's Dilemma the boost is larger for human partners. The figure places beliefs on the cooperation scale for comparison; the interaction is tested on the underlying 1–5 ratings (Table A.9). Error bars show 95% CIs.

Table A.8
Mixed-effects model: share of high cooperation beliefs (rating of 4 or 5, PD + SH stacked) on partner type, communication, and game type, with a random intercept and slope per participant (main study).

	Dep. var.: Share of high beliefs (rating of 4 or 5)				
	Coef.	**SE**	z	p	**95% CI**
Baseline (reference group)	0.5171	0.0430	12.01	<.001***	[0.433, 0.602]
Partner type (1 = AI, 0 = Human)	0.0123	0.0620	0.20	0.843	[-0.109, 0.134]
Communication (1 = yes, 0 = no)	0.2001	0.0607	3.30	<.001***	[0.081, 0.319]
Game type (1 = Stag Hunt, 0 = Prisoner's Dilemma)	0.1111	0.0385	2.89	0.004**	[0.036, 0.187]
Partner type × Communication	-0.0574	0.0911	-0.63	0.528	[-0.236, 0.121]
Partner type × Game type	-0.0513	0.0555	-0.92	0.355	[-0.160, 0.058]
Communication × Game type	-0.0166	0.0543	-0.31	0.760	[-0.123, 0.090]
Partner type × Comm × Game (three-way)	0.1029	0.0815	1.26	0.207	[-0.057, 0.263]
Comprehension check passed (centered control)	-0.0352	0.0587	-0.60	0.549	[-0.150, 0.080]
Observations	918				
Participants	459				
σ^2_u (random intercept)	0.1551				
σ^2_ε (residual)	0.0779				
ICC	0.666				

$^{***}p < .001$ $^{**}p < .01$ $^{*}p < .05$ $^{\dagger}p < .10$

Table A.9
Mixed-effects model: cooperation belief (raw 1–5 rating, PD + SH stacked) on partner type, communication, and game type, with a random intercept and slope per participant (main study).

	Dep. var.: Raw belief rating (1–5)				
	Coef.	**SE**	z	p	**95% CI**
Baseline (reference group)	3.3105	0.0838	39.49	<.001***	[3.146, 3.475]
Partner type (1 = AI, 0 = Human)	0.1506	0.1208	1.25	0.213	[-0.086, 0.387]
Communication (1 = yes, 0 = no)	0.5798	0.1182	4.90	<.001***	[0.348, 0.811]
Game type (1 = Stag Hunt, 0 = Prisoner's Dilemma)	0.4206	0.0844	4.99	<.001***	[0.255, 0.586]
Partner type × Communication	-0.3572	0.1774	-2.01	0.044*	[-0.705, -0.010]
Partner type × Game type	-0.2753	0.1216	-2.27	0.024*	[-0.514, -0.037]
Communication × Game type	-0.1529	0.1191	-1.28	0.199	[-0.386, 0.080]
Partner type × Comm × Game (three-way)	0.4346	0.1786	2.43	0.015*	[0.085, 0.785]
Comprehension check passed (centered control)	-0.0294	0.1150	-0.26	0.798	[-0.255, 0.196]
Observations	918				
Participants	459				
σ^2_u (random intercept)	0.5306				
σ^2_ε (residual)	0.3531				
ICC	0.601				

$^{***}p < .001$ $^{**}p < .01$ $^{*}p < .05$ $^{\dagger}p < .10$

Table A.10
Covariate-adjusted mixed-effects LPM: cooperation with demographic controls (main study).

	Dep. var.: Cooperation (binary)				
	Coef.	**SE**	z	p	**95% CI**
Baseline (reference group)	0.4241	0.1315	3.22	0.001**	[0.166, 0.682]
Partner type (1 = AI, 0 = Human)	-0.0862	0.0614	-1.41	0.160	[-0.206, 0.034]
Communication (1 = yes, 0 = no)	0.3058	0.0608	5.03	<.001***	[0.187, 0.425]
Partner type × Communication	-0.0289	0.0884	-0.33	0.744	[-0.202, 0.144]
Game type (1 = Stag Hunt, 0 = Prisoner's Dilemma)	0.2283	0.0439	5.20	<.001***	[0.142, 0.314]
Partner type × Game type	0.0195	0.0634	0.31	0.758	[-0.105, 0.144]
Communication × Game type	-0.1181	0.0621	-1.90	0.057†	[-0.240, 0.004]
Partner type × Comm × Game (three-way)	0.1287	0.0933	1.38	0.168	[-0.054, 0.311]
Age (years)	0.0018	0.0013	1.46	0.144	[-0.001, 0.004]
Female (1 = yes)	0.0914	0.0342	2.67	0.007**	[0.024, 0.158]
College degree+ (1 = yes)	-0.0767	0.0354	-2.17	0.030*	[-0.146, -0.007]
Ideology (1 = lib, 7 = cons)	-0.0082	0.0095	-0.86	0.389	[-0.027, 0.011]
AI awareness (1–5)	0.0161	0.0192	0.84	0.401	[-0.021, 0.054]
AI usage (mean of 3 items, 1–5)	-0.0313	0.0187	-1.68	0.094†	[-0.068, 0.005]
Log completion time	-0.0025	0.0470	-0.05	0.957	[-0.095, 0.090]
Observations	920				
Participants	460				
σ^2_u (random intercept)	0.1216				
σ^2_ε (residual)	0.0954				
ICC	0.560				

$^{***}p < .001$ $^{**}p < .01$ $^{*}p < .05$ $^{\dagger}p < .10$

Table A.11
Covariate-adjusted OLS regressions of non-cooperation dependent variables on the 2×2 factorial structure with demographic controls (main study).

	(1) *Trust*	(2) *Agency*	(3) *Honest/Fair*	(4) *Belief (PD)*	(5) *Belief (SH)*	(6) *Regulation*
Baseline (reference group)	2.488***	3.896***	2.856***	2.401***	3.577***	5.092***
	(0.265)	(0.384)	(0.235)	(0.327)	(0.326)	(0.488)
Partner type (1 = AI, 0 = Human)	-0.120	-1.680***	-0.015	0.093	-0.121	0.168
	(0.100)	(0.145)	(0.089)	(0.124)	(0.123)	(0.184)
Communication (1 = yes, 0 = no)	0.377***	-0.021	0.333***	0.530***	0.448***	0.380*
	(0.100)	(0.144)	(0.088)	(0.123)	(0.122)	(0.183)
Partner type × Communication	-0.208	0.193	-0.344**	-0.297†	0.083	-0.123
	(0.142)	(0.206)	(0.126)	(0.176)	(0.175)	(0.262)
Age (years)	0.004	-0.009*	0.002	0.011**	0.007*	0.013**
	(0.003)	(0.004)	(0.002)	(0.003)	(0.003)	(0.005)
Female (1 = yes)	0.109	-0.087	0.108†	0.222*	0.222*	0.410**
	(0.070)	(0.102)	(0.062)	(0.087)	(0.087)	(0.130)
College degree+ (1 = yes)	0.053	-0.162	-0.191**	-0.043	-0.015	-0.052
	(0.073)	(0.106)	(0.065)	(0.090)	(0.090)	(0.134)
Ideology (1 = lib, 7 = cons)	-0.033†	0.011	-0.036*	-0.041†	-0.025	-0.187***
	(0.020)	(0.028)	(0.017)	(0.024)	(0.024)	(0.036)
AI awareness (1–5)	-0.065	-0.051	-0.096**	0.017	-0.026	0.019
	(0.040)	(0.057)	(0.035)	(0.049)	(0.049)	(0.073)
AI usage (mean of 3 items, 1–5)	0.159***	0.144**	0.143***	0.083†	0.035	-0.121†
	(0.038)	(0.056)	(0.034)	(0.047)	(0.047)	(0.071)
Log completion time	0.225*	0.207	0.161†	0.117	-0.095	-0.117
	(0.097)	(0.140)	(0.086)	(0.120)	(0.119)	(0.178)
Observations	459	459	459	459	459	460
R^2	0.127	0.364	0.124	0.109	0.091	0.121

$^{***}p < .001$ $^{**}p < .01$ $^{*}p < .05$ $^{\dagger}p < .10$. Standard errors in parentheses.

Table A.12
Covariate-adjusted mixed-effects model: cooperation beliefs (PD + SH stacked) with controls (main study).

	Dep. var.: Cooperation belief (1–5)				
	Coef.	**SE**	z	p	**95% CI**
Baseline (reference group)	2.7787	0.2829	9.82	<.001***	[2.224, 3.333]
Partner type (1 = AI, 0 = Human)	0.1233	0.1219	1.01	0.312	[-0.116, 0.362]
Communication (1 = yes, 0 = no)	0.5651	0.1209	4.67	<.001***	[0.328, 0.802]
Partner type × Communication	-0.3240	0.1746	-1.86	0.064†	[-0.666, 0.018]
Game type (1 = Stag Hunt, 0 = Prisoner's Dilemma)	0.4206	0.0844	4.99	<.001***	[0.255, 0.586]
Partner type × Game type	-0.2753	0.1216	-2.27	0.024*	[-0.514, -0.037]
Communication × Game type	-0.1529	0.1191	-1.28	0.199	[-0.386, 0.080]
Partner type × Comm × Game (three-way)	0.4346	0.1786	2.43	0.015*	[0.085, 0.785]
Age (years)	0.0088	0.0027	3.23	0.001**	[0.004, 0.014]
Female (1 = yes)	0.2223	0.0744	2.99	0.003**	[0.076, 0.368]
College degree+ (1 = yes)	-0.0287	0.0770	-0.37	0.709	[-0.180, 0.122]
Ideology (1 = lib, 7 = cons)	-0.0331	0.0207	-1.60	0.110	[-0.074, 0.007]
AI awareness (1–5)	-0.0043	0.0418	-0.10	0.918	[-0.086, 0.078]
AI usage (mean of 3 items, 1–5)	0.0588	0.0405	1.45	0.146	[-0.021, 0.138]
Log completion time	0.0111	0.1022	0.11	0.914	[-0.189, 0.211]
Observations	918				
Participants	459				
σ^2_u (random intercept)	0.3942				
σ^2_ε (residual)	0.4483				
ICC	0.468				

$^{***}p < .001$ $^{**}p < .01$ $^{*}p < .05$ $^{\dagger}p < .10$

Table A.13
Covariate-adjusted logistic regression of cooperation with demographic controls, standard errors clustered on the participant (main study).

	Dep. var.: Cooperation (binary)				
	Coef.	**SE**	z	p	**95% CI**
Baseline (reference group)	-0.5577	0.6636	-0.84	0.401	[-1.858, 0.743]
Partner type (1 = AI, 0 = Human)	-0.3909	0.2843	-1.37	0.169	[-0.948, 0.166]
Communication (1 = yes, 0 = no)	1.3642	0.2908	4.69	<.001***	[0.794, 1.934]
Partner type × Communication	-0.1712	0.4140	-0.41	0.679	[-0.983, 0.640]
Game type (1 = Stag Hunt, 0 = Prisoner's Dilemma)	0.9870	0.2090	4.72	<.001***	[0.577, 1.397]
Partner type × Game type	0.0658	0.3040	0.22	0.829	[-0.530, 0.662]
Communication × Game type	-0.2325	0.3169	-0.73	0.463	[-0.854, 0.389]
Partner type × Comm × Game (three-way)	0.8420	0.5588	1.51	0.132	[-0.253, 1.937]
Age (years)	0.0129	0.0065	1.96	0.050*	[0.000, 0.026]
Female (1 = yes)	0.4593	0.1800	2.55	0.011*	[0.106, 0.812]
College degree+ (1 = yes)	-0.4330	0.1852	-2.34	0.019*	[-0.796, -0.070]
Ideology (1 = lib, 7 = cons)	-0.0385	0.0518	-0.74	0.457	[-0.140, 0.063]
AI awareness (1–5)	0.1016	0.1025	0.99	0.322	[-0.099, 0.302]
AI usage (mean of 3 items, 1–5)	-0.1859	0.0984	-1.89	0.059†	[-0.379, 0.007]
Log completion time	0.0280	0.2434	0.11	0.908	[-0.449, 0.505]
Observations	920				
Clusters (participants)	460				
Pseudo-R^2 (McFadden)	0.133				

$^{***}p < .001$ $^{**}p < .01$ $^{*}p < .05$ $^{\dagger}p < .10$

Coefficients are on the log-odds scale and are therefore not directly comparable in magnitude to the percentage-point estimates of the linear probability model in Table A.10; the comparison of interest is the sign and significance of each term. Standard errors are clustered on the participant, which absorbs the same within-person dependence across the two games as the random intercept in the LPM.

B. Additional Survey Items

Survey items were presented after the game portion of the study. Items are grouped by domain. Response options are shown in italics below each question.

B.1. AI Attitudes

1. Overall, when thinking about artificial intelligence (AI), would you say you are more excited or concerned about the growth in artificial intelligence?
   Mostly Excited / Somewhat Excited / Neutral / Somewhat Concerned / Mostly Concerned
2. Some people say that companies should be free to develop AI however they see fit with minimal to no government regulation. Others say that the government has a crucial role in regulating this new technology. On a 1 to 7 scale, with 1 indicating no regulation, and 7 indicating the most stringent regulation possible, where do you place yourself?
   1 – No Regulation 2 3 4 5 6 7 – Full Regulation
3. To what extent would you support or oppose having a federal agency regulate the use of artificial intelligence?
   Strongly Oppose / Somewhat Oppose / Neutral / Somewhat Support / Strongly Support
4. How likely do you think it is that an AI could accidentally cause a catastrophic event?
   Very Unlikely / Somewhat Unlikely / Somewhat Likely / Very Likely

B.2. ChatGPT Familiarity and Usage

1. How much, if anything, have you heard about ChatGPT, an artificial intelligence (AI) large language model (LLM) program used to understand and create text?
   A great deal / A lot / A moderate amount / A little / None at all
2. How often, if at all, do you use large language models such as ChatGPT for entertainment?
   Very often / Fairly often / Occasionally / Rarely / I have never used a large language model for entertainment
3. How often, if at all, do you use large language models such as ChatGPT for learning purposes?
   Very often / Fairly often / Occasionally / Rarely / I have never used a large language model for learning purposes
4. How often, if at all, do you use large language models such as ChatGPT for work purposes?
   Very often / Fairly often / Occasionally / Rarely / I have never used a large language model for work purposes

B.3. Demographics

1. What is your current age in years?
   {Free integer field}

2. What is your gender?

   Male / Female / Other/Prefer not to say

3. What racial or ethnic group describes you? If more than one, please tell us the group you most identify with and/or one that best describes you.

   Asian / Black or African American / Hispanic or Latino/a / Middle Eastern or North African / Native American / Native Hawaiian or Other Pacific Islander / White / Other

4. What is the highest level of school you have completed or the highest degree you have received?

   Elementary school / High school/Equivalent / Vocational/Technical school (2 year) / Some college / College graduate (4 year) / Master's Degree (MS) / Doctoral Degree (PhD) / Professional Degree (MD, JD, etc.) / Other

5. Generally speaking, do you think of yourself as a Republican, a Democrat, an Independent, or other?

   Republican / Democrat / Independent / Other

   If Republican/Democrat: Would you consider yourself a strong [Republican/Democrat] or a not very strong [Republican/Democrat]?

   Strong [Republican/Democrat] / Not very strong [Republican/Democrat]

6. We hear a lot of talk these days about liberals and conservatives. Here is a 7-point scale on which the political views people might hold are arranged. Where would you place yourself on this scale?

   Extremely liberal / Liberal / Slightly liberal / Moderate; middle of the road / Slightly conservative / Conservative / Extremely conservative

B.4. Game-Specific Belief Items

1. How likely do you think it is that the other [player/player's AI bot] chose B in Game 1?

   Extremely Likely / Somewhat Likely / Neither Likely nor Unlikely / Somewhat Unlikely / Extremely Unlikely

2. How likely do you think it is that the other [player/player's AI bot] chose B in Game 2?

   Extremely Likely / Somewhat Likely / Neither Likely nor Unlikely / Somewhat Unlikely / Extremely Unlikely

3. How trustworthy do you find the other [player/player's AI bot]?

   Very Trustworthy / Somewhat Trustworthy / Neither Trustworthy nor Untrustworthy / Somewhat Untrustworthy / Very Untrustworthy

B.5. Agentic and Honest/Fair Scale

Please indicate the extent to which you agree or disagree with the following statements about the other player's AI bot behavior:

Table B.1
Agentic and honest/fair scale survey items.

Statement	Strongly Agree	Somewhat Agree	Neither	Somewhat Disagree	Strongly Disagree
Has intentions	○	○	○	○	○
Has a mind of its/their own	○	○	○	○	○
Is honest	○	○	○	○	○
Is selfish	○	○	○	○	○
Is sincere	○	○	○	○	○
Is unbiased	○	○	○	○	○

C. Effective Sample Size Under the Strategy Method

The pilot study uses the strategy method: participants in communication conditions specify a contingent choice for each of seven possible messages the counterpart could send, while participants in no-communication conditions make a single unconditional choice. This creates an unbalanced panel in which each participant contributes either 1 or 7 binary observations to the mixed-effects model. Because the multiple observations from the same participant are correlated, the nominal observation count ($N_{\text{obs}} = 1{,}223$) overstates the amount of independent information in the data. We compute the effective sample size to quantify the true statistical precision.

C.1. Intraclass Correlation Coefficient

The mixed-effects linear probability model (Table A.2) includes a random intercept $u_i \sim \mathcal{N}(0, \sigma_u^2)$ for each participant. The intraclass correlation coefficient (ICC) measures the proportion of total variance attributable to between-participant differences:

$$\text{ICC} = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\varepsilon^2} \tag{C.1}$$

where σ_ε^2 is the residual (within-participant) variance. From our model, $\sigma_u^2 = 0.0854$ and $\sigma_\varepsilon^2 = 0.1456$, yielding $\text{ICC} = 0.370$. That is, 37% of the variance in cooperation decisions is between participants and 63% is within participants (across the seven contingencies).

C.2. Design Effect

The design effect (DEFF) quantifies how much the clustering inflates the variance of parameter estimates relative to a simple random sample of the same size. For a cluster of size m_i:

$$\text{DEFF}_i = 1 + (m_i - 1) \times \text{ICC} \tag{C.2}$$

Table C.1
Effective sample size by condition (pilot study). Design effect computed from ICC = 0.370.

Condition	n	Obs/person	Total obs	DEFF	N_{eff}
HH_NOCOM	73	1	73	1.00	73
HH_wCOM	80	7	560	3.22	174
AIH_NOCOM	72	1	72	1.00	72
AIH_wCOM	74	7	518	3.22	161
Total	299	4.09	1223	2.14	571

Kish effective $N = 414$ (accounts for unbalanced cluster sizes).

For no-communication participants ($m_i = 1$), DEFF $= 1$ — there is no clustering penalty. For communication participants ($m_i = 7$), DEFF $= 1 + 6 \times 0.370 = 3.22$ — each participant's seven choices carry the statistical weight of approximately $7/3.22 \approx 2.2$ independent observations.

C.3. Effective Sample Size

The effective sample size adjusts the nominal observation count by the design effect. For balanced clusters, this is simply $N_{\text{eff}} = N_{\text{obs}}/\text{DEFF}$. Because our clusters are unbalanced (sizes 1 and 7), we use the Kish (1965) formula:

$$N_{\text{eff}} = \frac{\left(\sum_{i=1}^{K} m_i\right)^2}{\sum_{i=1}^{K} m_i \times \text{DEFF}_i} \tag{C.3}$$

where $K = 299$ is the number of participants and m_i is the cluster size for participant i.

The 1,223 nominal observations carry the statistical precision of approximately 414 independent observations (Kish N_{eff}), or about 1.38 times the 299 participants. The design effect is driven entirely by the communication conditions; the no-communication conditions contribute observations at full efficiency.

C.4. Main Study: Stacked Panel (PD + SH)

In the main study, each participant contributes two binary observations—one Prisoner's Dilemma decision and one Stag Hunt decision—yielding a balanced panel with uniform cluster size $m = 2$. The hierarchical mixed-effects model includes both a random intercept and a random slope for game type. The ICC based on the random intercept is substantially higher than in the pilot study (ICC = .573 vs. .370), reflecting the strong within-person consistency in cooperation tendency across games: participants who cooperate in the PD tend to also cooperate in the SH.

With $m = 2$ and ICC = .573:

$$\text{DEFF} = 1 + (2 - 1) \times .573 = 1.573 \tag{C.4}$$

The 920 observations carry the precision of approximately 585 independent observations, or 1.27 times the 460 participants. Although the ICC is higher than in the pilot study, the design effect is more modest because the cluster

Table C.2
Effective sample size by condition (main study). Design effect computed from ICC = 0.573. Each participant contributes two observations (Prisoner's Dilemma and Stag Hunt).

Condition	n	Obs/person	Total obs	DEFF	N_{eff}
Human No Comm	127	2	254	1.57	162
Human Comm	127	2	254	1.57	162
AI No Comm	117	2	234	1.57	149
AI Comm	89	2	178	1.57	113
Total	460	2	920	1.57	585

ICC based on random intercept only; random slope for game type provides additional structure.

size is small ($m = 2$). The random slope for game type captures additional within-person structure—the PD-vs-SH difference varies across participants—but this is not reflected in the simple design-effect formula, which conditions only on the random intercept. The effective N should thus be interpreted as a conservative lower bound.

D. Pre-Registration: Pilot Study

The pilot study was pre-registered at OSF prior to data collection (registration ID: `[ID withheld]`; [OSF link withheld for blind review]; registered August 21, 2023). Below we summarize the key elements and note deviations from the registered plan.

D.1. Design

The registration described a 2 × 2 factorial between-subjects design crossing partner type (human vs. AI bot) with communication (present vs. absent) in a one-shot Prisoner's Dilemma. Two additional delegator conditions were included so that no deception was involved: participants whose AI bot played on their behalf did not make binding choices but were asked to predict the players' behavior. Randomization was conducted via Qualtrics.

D.2. Sample

Registered plan: 100 participants per condition, 600 total, recruited from Amazon Mechanical Turk (U.S., English-speaking, 18+). Participation fee: $1, plus $0.25–$1 bonus.

Stopping rule: Data collection would stop at 600 total participants, even if individual conditions had not reached 100.

Actual: 640 responses collected; 597 completed after removing previews and invalid conditions. Final analytic sample: 447 after comprehension exclusion (see below).

D.3. Hypotheses

Four hypotheses were pre-registered:

1. Cooperation rates will be lower in the human–AI conditions compared to the human–human conditions.

2. Cooperation rates will be higher in the human–human condition with communication than without.
3. Cooperation rates will be *lower* in the human–AI condition with communication than without communication.
4. Support for AI regulation will be higher in human–AI conditions than in human–human conditions.

Note that H3 predicted that communication would *decrease* cooperation with AI partners—the opposite of what was observed. This prediction was based on the possibility that communication with a known AI bot might feel hollow or manipulative, thereby reducing cooperation.

D.4. Statistical Models

Registered: Simple *t*-tests and non-parametric Mann–Whitney tests; logistic regression with demographic covariates. Two-tailed tests at significance levels of both .05 and .10.

Actual: We report *t*-tests and Mann–Whitney tests as registered. We additionally report a mixed-effects linear probability model with participant random intercepts to properly handle the strategy method's clustered structure (see Section C). Logistic regression with demographic covariates was not conducted but can be added as a robustness check.

Justification: The strategy method creates 7 correlated binary observations per participant in communication conditions, violating the independence assumption of simple *t*-tests. The mixed-effects model accounts for this clustering while retaining all individual-level variation.

D.5. Deviations from Registration

1. **Comprehension check used as exclusion, not covariate.** The registration stated: "We included one comprehension question, but this will be used as a covariate, not as an exclusion criteria." In practice, 150 of 597 participants (25.1%) failed the comprehension check and were excluded. This is a substantive deviation. We excluded rather than covaried because including participants who demonstrably misunderstood the payoff structure would introduce noise that a covariate cannot fully address. The correct comprehension answer ("25 cents") requires understanding the payoff matrix; participants who answered "$1" likely confused their own payoff with their partner's, calling into question whether their choices reflected the intended incentive structure.
2. **Mixed-effects model added.** The registration specified *t*-tests and Mann–Whitney tests. We report these as registered but supplement them with a mixed-effects model that properly handles the strategy method's within-participant clustering. This is a methodological improvement, not a change in hypothesis or inference direction. The naive *t*-tests and the mixed model yield qualitatively identical conclusions.
3. **Logistic regression with covariates not reported.** The registration mentioned logistic regression with demographic covariates. This analysis was not conducted for the current manuscript but can be provided upon request.

4. **Dual significance thresholds.** The registration specified reporting at both $\alpha = .05$ and $\alpha = .10$. We report exact p-values throughout, which subsumes both thresholds.

E. Formal Hypothesis Specifications

This appendix provides the formal notation and directional inequalities for each pre-registered hypothesis. See Section 3 for the plain-language statements and theoretical motivation.

E.1. Notation

We use two-letter abbreviations matching the figure labels throughout the paper:

We define the following dependent variables:

- $P(Game_{\text{Label}})$: mean cooperation rate (Choice B) in game $Game \in \{PD, SH\}$ for a given treatment condition. Values near 1 indicate cooperation/Stag; values near 0 indicate defection/Hare.
- *Trust*(Label): mean trust rating of the other player, on a 1–5 scale (1 = Very Untrustworthy, 5 = Very Trustworthy).
- *Agency*(Label): mean score on the agentic scale, combining two items: "has intentions" and "has a mind of their own."
- *AIReg*(Label): mean support for AI regulation, on a 1–7 scale (1 = No Regulation, 7 = Full Regulation).

For aggregate comparisons across communication conditions, we use:

- HH pooled: HH, No Com & HH, Com.
- AI pooled: AI, No Com & AI, Com.

E.2. H1 — AI Partner Reduces Cooperation

H1a: $P(PD_{\text{AI, No Com}}) < P(PD_{\text{HH, No Com}})$

H1b: $P(PD_{\text{AI, Com}}) < P(PD_{\text{HH, Com}})$

H1c: $P(SH_{\text{AI, No Com}}) < P(SH_{\text{HH, No Com}})$

H1d: $P(SH_{\text{AI, Com}}) < P(SH_{\text{HH, Com}})$

E.3. H2 — Communication Increases Cooperation

H2a: $P(PD_{\text{HH, No Com}}) < P(PD_{\text{HH, Com}})$

H2b: $P(PD_{\text{AI, No Com}}) < P(PD_{\text{AI, Com}})$

H2c: $P(SH_{\text{HH, No Com}}) < P(SH_{\text{HH, Com}})$

H2d: $P(SH_{\text{AI, No Com}}) < P(SH_{\text{AI, Com}})$

Table E.1
Treatment condition key

Label	Description	Partner Type	Communication
HH, No Com	Human–Human, No Communication	Human	No
HH, Com	Human–Human, With Communication	Human	Yes
AI, No Com	Human–AI, No Communication	AI	No
AI, Com	Human–AI, With Communication	AI	Yes

E.4. H3 — Communication Effect Is Larger with AI Partner

H3a: $\left[P(PD_{\text{HH, Com}}) - P(PD_{\text{HH, No Com}})\right] < \left[P(PD_{\text{AI, Com}}) - P(PD_{\text{AI, No Com}})\right]$

H3b: $\left[P(SH_{\text{HH, Com}}) - P(SH_{\text{HH, No Com}})\right] < \left[P(SH_{\text{AI, Com}}) - P(SH_{\text{AI, No Com}})\right]$

E.5. H4 — Lower Trust with AI Partner

$$\textit{Trust}(\text{AI pooled}) < \textit{Trust}(\text{HH pooled})$$

E.6. H5 — Communication Increases Trust

H5a: $\textit{Trust}(\text{HH, No Com}) < \textit{Trust}(\text{HH, Com})$

H5b: $\textit{Trust}(\text{AI, No Com}) < \textit{Trust}(\text{AI, Com})$

E.7. H6 — Trust Gap in Communication Conditions

$$\textit{Trust}(\text{AI, Com}) < \textit{Trust}(\text{HH, Com})$$

E.8. H7 — Higher Agency Attribution to Humans

H7a: $\textit{Agency}(\text{AI, No Com}) < \textit{Agency}(\text{HH, No Com})$

H7b: $\textit{Agency}(\text{AI, Com}) < \textit{Agency}(\text{HH, Com})$

E.9. H8 — AI Exposure Increases Support for Regulation

H8a: $\textit{AIReg}(\text{HH, No Com}) < \textit{AIReg}(\text{AI, No Com})$

H8b: $\textit{AIReg}(\text{HH, Com}) < \textit{AIReg}(\text{AI, Com})$

F. Statistical Methods

This appendix provides technical details on the statistical procedures used in the main analysis. All analyses were conducted in Python using `pingouin` (ANOVA, t-tests, Tukey HSD, Cohen's d, χ^2 and Mann–Whitney tests, Cronbach's α), `statsmodels` (OLS, logistic, and mixed-effects models, their contrasts, and confidence intervals), and `scipy` (Fisher's exact test).

F.1. One-Way ANOVA and Tukey HSD

For each dependent variable measured at the treatment level (cooperation, trust, perceived agency, honesty/fairness, cooperation beliefs, regulation support), we first conducted a one-way between-subjects ANOVA with the four experimental conditions (HH No Com, HH Com, AI No Com, AI Com) as the factor. The ANOVA tests the omnibus null hypothesis that all group means are equal:

$$H_0 : \mu_{\text{HH,NoCom}} = \mu_{\text{HH,Com}} = \mu_{\text{AI,NoCom}} = \mu_{\text{AI,Com}}$$

We report the F-statistic, p-value, and partial eta-squared (η_p^2) as a measure of effect size.

When the omnibus F-test was significant, we followed up with Tukey's Honestly Significant Difference (HSD) test for all pairwise comparisons.[8] The procedure controls the family-wise error rate across the six possible pairwise comparisons by comparing each observed mean difference against the studentized range distribution. Effect sizes for pairwise contrasts are reported as Cohen's d.

For two-group comparisons (e.g., pooled partner-type effects), we used Welch's independent-samples t-test, which does not assume equal variances, and report Cohen's d.

Assumptions. The ANOVA assumes independence of observations (ensured by random assignment), approximate normality of residuals within groups, and homogeneity of variances across groups. With group sizes of 89–127 participants per condition and no extreme skewness, the ANOVA F-test is robust to moderate violations of normality Lix, Keselman and Keselman (1996). Because the four conditions have slightly unequal sample sizes after exclusions, we use the Tukey–Kramer procedure Kramer (1956), which generalizes Tukey HSD by replacing the common n in the standard error with $\sqrt{MS_W \cdot \frac{1}{2}(1/n_i + 1/n_j)}$ for each pair (i, j). The Tukey–Kramer method controls the family-wise error rate at the nominal level for arbitrary group sizes and is conservative (i.e., the true Type I error rate is at most α).

F.2. Mixed-Effects Linear Probability Model

The primary cooperation analysis uses a mixed-effects linear probability model (LPM) to jointly estimate the effects of partner type, communication, and game type while accounting for the within-subject dependence introduced by each participant playing both the Prisoner's Dilemma and Stag Hunt.

[8] For each pair of groups (i, j), the Tukey–Kramer procedure computes $q_{ij} = |\bar{y}_i - \bar{y}_j| / \sqrt{MS_W \cdot \frac{1}{2}(1/n_i + 1/n_j)}$, where MS_W is the within-group mean square from the ANOVA and n_i, n_j are the respective group sizes. This statistic is compared against the critical value of the studentized range distribution $q_{\alpha,k,N-k}$, where k is the number of groups and N the total sample size. Because the critical value is calibrated to the distribution of the maximum pairwise difference, simultaneous coverage is maintained at the nominal α level across all $\binom{k}{2}$ comparisons without further multiplicity adjustment.

F.2.1. Model Specification

Predictor variables. All regression models use three binary indicators that encode the $2 \times 2 \times 2$ factorial structure. Throughout the regression tables in Appendix A, these appear under the following labels:

- **Partner type** ($\in \{0, 1\}$): coded 1 if the participant's counterpart was an AI bot, 0 if human. This is a between-subjects factor.
- **Communication** ($\in \{0, 1\}$): coded 1 if the condition included a pre-play communication stage, 0 otherwise. This is a between-subjects factor.
- **Game type** ($\in \{0, 1\}$): coded 1 for Stag Hunt, 0 for Prisoner's Dilemma. This is a within-subjects factor (each participant plays both games).

The reference category (all indicators at 0) is the Human–No Communication–Prisoner's Dilemma condition, so the intercept β_0 estimates the cooperation rate in that cell.

Each participant i contributes two binary observations: one for the Prisoner's Dilemma ($j = \text{PD}$) and one for the Stag Hunt ($j = \text{SH}$). The outcome $y_{ij} \in \{0, 1\}$ indicates whether the participant cooperated (chose B) in game j. The model is:

$$\begin{aligned} y_{ij} = \beta_0 &+ \beta_1 \,\text{PartnerType}_i + \beta_2 \,\text{Comm}_i + \beta_3 \,\text{GameType}_j \\ &+ \beta_4 \,(\text{PartnerType}_i \times \text{Comm}_i) \\ &+ \beta_5 \,(\text{PartnerType}_i \times \text{GameType}_j) \\ &+ \beta_6 \,(\text{Comm}_i \times \text{GameType}_j) \\ &+ \beta_7 \,(\text{PartnerType}_i \times \text{Comm}_i \times \text{GameType}_j) \\ &+ u_i + v_i \,\text{GameType}_j + \varepsilon_{ij} \end{aligned} \tag{F.1}$$

where the random effects are:

- $u_i \sim N(0, \sigma_u^2)$: random intercept capturing participant i's baseline cooperation tendency.
- $v_i \sim N(0, \sigma_v^2)$: random slope capturing participant i's differential response to game type.
- $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$: residual error.

The random intercept u_i absorbs stable individual differences in cooperativeness (e.g., some participants cooperate more regardless of condition), while the random slope v_i allows the PD-versus-SH gap to vary across participants (e.g.,

some participants are more sensitive to the game's payoff structure than others). The random effects (u_i, v_i) are assumed to follow a bivariate normal distribution with an unstructured covariance matrix, estimated via restricted maximum likelihood (REML).

F.2.2. Interpretation of Coefficients

In the LPM, each coefficient β_k has a direct interpretation as the change in the probability of cooperation associated with a one-unit change in the corresponding predictor. For example, $\beta_2 = 0.29$ means that communication increases the probability of cooperation by approximately 29 percentage points, holding other predictors constant. Interaction terms capture how the effect of one factor depends on the level of another.

F.2.3. Why a Linear Probability Model?

Although cooperation is binary, we use a linear model rather than a logistic (or probit) specification for three reasons. First, the LPM provides directly interpretable marginal effects without the need for post-estimation transformations. Second, with cooperation rates in the 40–90% range (far from the boundaries), the LPM approximation is accurate and the risk of predicted probabilities outside $[0, 1]$ is minimal. Third, the LPM facilitates straightforward comparison of interaction effects, which are notoriously difficult to interpret in nonlinear models Ai and Norton (2003). As a robustness check, we also estimated the covariate-adjusted cooperation model as a logistic regression with standard errors clustered on the participant, reported in Appendix A (Table A.13). Every treatment term agrees with the linear model in sign and significance, so the substantive conclusions do not depend on the linear specification.

F.2.4. Intraclass Correlation

The intraclass correlation coefficient (ICC) quantifies the proportion of total variance attributable to between-participant differences:

$$\text{ICC} = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\varepsilon^2}$$

A high ICC indicates that participants' cooperation decisions in the two games are positively correlated, justifying the use of random effects rather than treating all observations as independent.

F.3. OLS Factorial Regressions

For dependent variables measured once per participant (trust, perceived agency, honesty/fairness, regulation support), we estimated OLS regressions of the form:

$$y_i = \beta_0 + \beta_1 \,\text{PartnerType}_i + \beta_2 \,\text{Comm}_i + \beta_3 \,(\text{PartnerType}_i \times \text{Comm}_i) + \varepsilon_i$$

These regressions decompose the omnibus ANOVA into the two main effects and their interaction, providing coefficient estimates and standard errors for each factor. The interaction term β_3 tests whether the effect of communication differs by partner type (or equivalently, whether the effect of partner type differs by communication condition).

F.4. Covariate-Adjusted Models

As a robustness check, we re-estimated all models with seven pre-treatment covariates: age, gender (female indicator), education (college degree or higher), political ideology (1–7 scale), AI awareness (1–5), AI usage (mean of three items, 1–5), and log-transformed completion time. These covariates were selected because they are measured independently of the experimental manipulation and could plausibly predict the dependent variables. Under successful randomization, adding pre-treatment covariates should not substantially change the treatment effect estimates, but may reduce residual variance and improve precision. The covariate-adjusted results are reported in Appendix A.

G. Images Provided to Participants of the GPT Prompting Process and Result

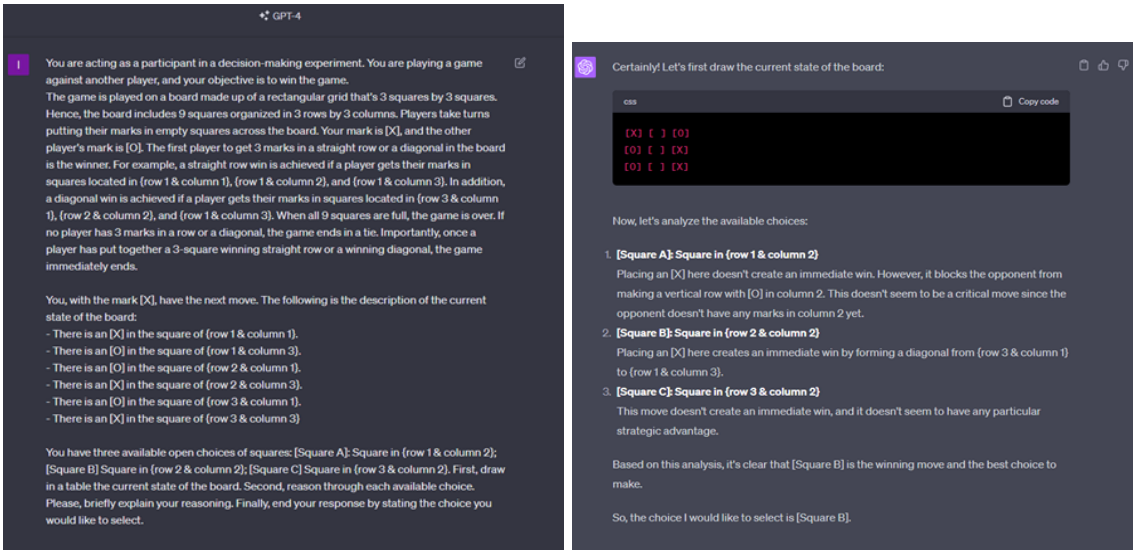


Figure G.1: Screenshots shown to participants demonstrating GPT's ability to play tic-tac-toe.

GIF: [URL redacted for blind review]

H. Data Collection Summary

Table H.1 summarises raw and analytic sample sizes, dropped cases, and data collection windows across studies and waves. The pilot study was a single Qualtrics deployment on MTurk Prime in August 2023. The main study comprises two waves: an initial MTurk Prime oTree run in October 2024 and a Connect (CloudResearch) replication conducted in March–May 2026; the analytic dataset pools the two.

Table H.1
Data collection summary across studies and waves, with per-filter dropout accounting. *Total* is the raw cohort entering the cleaning pipeline. Each row is attributed to the *first* filter that excludes it. *Incomplete* = did not finish (Qualtrics: not Finished/Preview/invalid Condition; oTree: did not reach `End` or `Debrief_Parent`). *AI-delegated* = oTree `is_bot=1` — humans in AIH conditions whose cooperation choices were generated by an AI agent acting on their behalf, and whose behavioral data is therefore non-human-authored. *No condition* = waiting-room timeout before treatment assignment. *No decision* = partner timed out mid-game, leaving PD or SH unrecorded. *Failed check* = pilot-study comprehension-check failure (no analogue in the oTree waves). *Valid* is the analytic sample. The main-study Connect total reflects the 290 participants recoverable from oTree exports; an additional 10 PIDs marked complete on the Connect platform are unrecoverable due to oTree-DB resets in early March 2026.

Study	Wave	Total	Incomp.	AI-deleg.	No cond.	No decis.	Failed chk	Valid	Collection dates
Pilot	Qualtrics (MTurk Prime)	640	43	—	—	—	150	447	Aug 15–22, 2023
Main	MTurk Prime (oTree)	542	68	155	0	0	—	319	Oct 24, 2024
	Connect / CloudResearch (oTree)	290	2	82	13	52	—	141	Mar 6 – May 8, 2026
	Pooled (analytic)	*832*	*70*	*237*	*13*	*52*	—	*460*	*Oct 24, 2024 – May 8, 2026*

I. Sample Characteristics

Table I.1 reports the sociodemographic composition of the two analytic samples. The *pilot-study sample* corresponds to the pilot study strategy-method experiment ($N = 447$, recruited via MTurk Prime and run through Qualtrics). The *main-study sample* corresponds to the merged synchronous oTree experiment ($N = 460$): the original MTurk Prime collection ($n = 319$) combined with the Connect/CloudResearch extension ($n = 141$); see Appendix M for collection details. Continuous variables are summarized by mean and standard deviation with observed range; categorical variables are reported as level frequencies with within-sample percentages. Sample-equivalence tests use Welch's t for continuous variables and Pearson's χ^2 for categorical variables; these test only between-sample differences and do not bear on the experimental analyses, which are within-sample.

The two samples are demographically similar on most observed characteristics. The principal exception is age: the main-study sample is older on average ($M = 45.8$ vs. $M = 41.5$ in the pilot-study sample; $p < .001$). Distributions of gender, race/ethnicity, education, political party, and political ideology do not differ significantly between samples. The AI-usage frequency index is reported only for the main-study sample because the pilot study measured LLM usage with binary yes/no items rather than the 5-point frequency scale used in the main study.

I.1. Within the Main-Study Sample: Comparison Across Recruitment Sources

The main-study sample pools two recruitment sources collected through the same oTree protocol: the original MTurk Prime wave ($n = 319$) and the Connect extension ($n = 141$). Table I.2 reports a within-main-study-sample diagnostic that contrasts the two sources on demographics (Panel A) and on the experimental dependent variables (Panel B). The two sources are statistically indistinguishable on age, gender, education, political party, political ideology, and—on the experimental DVs—on cooperation rates, trust, perceived agency, honesty/fairness, cooperation

Table I.1
Sociodemographic characteristics of the two analytic samples. Continuous variables: mean (SD) and observed range. Categorical variables: level frequencies as n (%). Sample-equivalence tests: Welch's t for continuous variables, Pearson χ^2 for categorical. Significance markers: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

Variable	Level/Statistic	**Pilot** (N=447)	**Main** (N=460)	Stat.	p
Age (years)	M (SD)	41.53 (11.85)	45.76 (14.02)	$t = -4.92$	$< .001^{***}$
	[min, max]	[22, 77]	[18, 86]		
Gender				$\chi^2 = 1.51$	.471
	Female	218 (48.8%)	242 (52.6%)		
	Male	225 (50.3%)	213 (46.3%)		
	Other/Prefer not to say	4 (0.9%)	5 (1.1%)		
Race/ethnicity				$\chi^2 = 3.41$	.637
	White	328 (73.4%)	339 (73.7%)		
	Black or African American	47 (10.5%)	38 (8.3%)		
	Hispanic or Latino/a	22 (4.9%)	33 (7.2%)		
	Asian	39 (8.7%)	39 (8.5%)		
	Native American	3 (0.7%)	2 (0.4%)		
	Other	8 (1.8%)	9 (2.0%)		
Education				$\chi^2 = 11.71$	.111
	Elementary school	2 (0.4%)	0 (0.0%)		
	High school or equivalent	44 (9.8%)	45 (9.8%)		
	Some college	98 (21.9%)	98 (21.3%)		
	Vocational/Technical school (2 year)	16 (3.6%)	21 (4.6%)		
	College graduate (4 year)	206 (46.1%)	213 (46.3%)		
	Master's Degree (MS)	71 (15.9%)	57 (12.4%)		
	Professional Degree (MD, JD, etc.)	6 (1.3%)	12 (2.6%)		
	Doctoral Degree (PhD)	4 (0.9%)	14 (3.0%)		
Political party				$\chi^2 = 1.47$	.688
	Democrat	219 (49.0%)	214 (46.5%)		
	Republican	108 (24.2%)	110 (23.9%)		
	Independent	110 (24.6%)	128 (27.8%)		
	Other	10 (2.2%)	8 (1.7%)		
Ideology (1=very liberal, 7=very conservative)	M (SD)	3.62 (1.91)	3.53 (1.82)	$t = 0.72$	.472
	[min, max]	[1, 7]	[1, 7]		
AI awareness (1–5)	M (SD)	3.36 (1.00)	3.62 (1.03)	$t = -3.88$	$< .001^{***}$
	[min, max]	[1, 5]	[1, 5]		
AI usage (1–5, mean of 3 frequency items)	M (SD)	—	2.58 (1.05)		—
	[min, max]		[1, 5]		

beliefs, and AI regulation support. They differ significantly on race/ethnicity ($\chi^2 = 17.46$, $p = .004$) and on self-reported AI awareness and AI usage frequency (both higher among Connect participants). The PD comprehension check was administered in both waves; the SH check was added for the Connect wave. Pass rates by wave are reported in Table I.3.

I.2. Comprehension-Check Pass Rates

Table I.3 reports comprehension-check pass rates for the main study analytic sample, broken out by wave (Panel A) and by treatment cell within each wave (Panel B). The PD comprehension item was administered in both the MTurk Prime ($n = 319$) and Connect ($n = 141$) waves, with a pooled pass rate of 87.8% (404/460). The SH comprehension item was added only in the Connect wave; among Connect participants the pass rate was 84.4% (119/141). Panel C shows the joint distribution of PD and SH passage in the Connect wave: 77.3% passed both checks, 11.3% passed PD only, 7.1% passed SH only, and 4.3% failed both. Comprehension failers are retained in the analytic dataset and flagged via `failed_comprehension_pd` (and `failed_comprehension_sh` in Connect); a sensitivity analysis restricting the sample to PD-passers ($N = 404$) confirms that the substantive findings are unchanged (see Appendix C or the accompanying sensitivity outputs).

Table I.2

Comparison within the main-study sample by recruitment source: oTree-MTurk Prime vs. Connect (CloudResearch). Continuous variables: mean (SD) and observed range. Categorical variables: n (%) per level. Binary cooperation rates: % cooperating (analytic n). Equivalence tests: Welch's t for continuous, Pearson χ^2 for categorical or 2×2 binary. Significance markers: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

Variable	Level/Statistic	MTurk Prime (n=319)	Connect (CloudResearch) (n=141)	Stat.	p
Panel A: Sociodemographics					
Age (years)	M (SD)	46.24 (13.02)	44.68 (16.06)	$t = 1.02$	.311
	[min, max]	[23, 86]	[18, 79]		
Gender				$\chi^2 = 0.28$	.870
	Female	168 (52.7%)	74 (52.5%)		
	Male	147 (46.1%)	66 (46.8%)		
	Other/Prefer not to say	4 (1.3%)	1 (0.7%)		
Race/ethnicity				$\chi^2 = 17.46$	.004**
	White	251 (78.7%)	88 (62.4%)		
	Black or African American	20 (6.3%)	18 (12.8%)		
	Hispanic or Latino/a	16 (5.0%)	17 (12.1%)		
	Asian	25 (7.8%)	14 (9.9%)		
	Native American	2 (0.6%)	0 (0.0%)		
	Other	5 (1.6%)	4 (2.8%)		
Education				$\chi^2 = 5.99$	.424
	High school or equivalent	28 (8.8%)	17 (12.1%)		
	Some college	62 (19.4%)	36 (25.5%)		
	Vocational/Technical school (2 year)	14 (4.4%)	7 (5.0%)		
	College graduate (4 year)	155 (48.6%)	58 (41.1%)		
	Master's Degree (MS)	39 (12.2%)	18 (12.8%)		
	Professional Degree (MD, JD, etc.)	9 (2.8%)	3 (2.1%)		
	Doctoral Degree (PhD)	12 (3.8%)	2 (1.4%)		
Political party				$\chi^2 = 6.16$	.104
	Democrat	152 (47.6%)	62 (44.0%)		
	Republican	78 (24.5%)	32 (22.7%)		
	Independent	81 (25.4%)	47 (33.3%)		
	Other	8 (2.5%)	0 (0.0%)		
Ideology (1=very liberal, 7=very conservative)	M (SD)	3.56 (1.80)	3.46 (1.87)	$t = 0.52$	.605
	[min, max]	[1, 7]	[1, 7]		
AI awareness (1–5)	M (SD)	3.50 (1.01)	3.89 (1.04)	$t = -3.68$	< .001***
	[min, max]	[1, 5]	[1, 5]		
AI usage (1–5, mean of 3 frequency items)	M (SD)	2.43 (1.04)	2.93 (0.99)	$t = -5.00$	< .001***
	[min, max]	[1, 5]	[1, 5]		
Panel B: Experimental DVs and behavioral responses					
PD cooperation rate	% (n)	56.4% (n=319)	51.8% (n=141)	$\chi^2 = 0.68$	.410
SH cooperation rate	% (n)	74.6% (n=319)	78.0% (n=141)	$\chi^2 = 0.44$	.505
Trust in other player (1–5)	M (SD)	3.40 (0.78)	3.34 (0.81)	$t = 0.76$	.450
	[min, max]	[1, 5]	[1, 5]		
Perceived agency (1–5)	M (SD)	3.28 (1.37)	3.32 (1.27)	$t = -0.33$	.739
	[min, max]	[1, 5]	[1, 5]		
Perceived honesty/fairness (1–5)	M (SD)	3.22 (0.68)	3.22 (0.73)	$t = 0.05$	.959
	[min, max]	[1.5, 5]	[1, 5]		
Belief other cooperated, PD (1–5)	M (SD)	3.59 (0.96)	3.56 (0.98)	$t = 0.36$	.721
	[min, max]	[1, 5]	[1, 5]		
Belief other cooperated, SH (1–5)	M (SD)	3.89 (0.94)	3.90 (0.97)	$t = -0.13$	.895
	[min, max]	[1, 5]	[1, 5]		
AI regulation support (1–7)	M (SD)	4.88 (1.44)	5.09 (1.46)	$t = -1.41$	.160
	[min, max]	[1, 7]	[1, 7]		
Comprehension passed, PD (%)	% (n)	87.5% (n=319)	88.7% (n=141)	$\chi^2 = 0.04$	.837
Comprehension passed, SH (%)	% (n)	—	84.4% (n=141)		—

Table I.3
Comprehension-check pass rates across main-study waves and treatment cells. The PD comprehension item was administered in both the MTurk Prime and Connect waves; the SH comprehension item was only added in the Connect wave (no SH check in MTurk). Pass rates are calculated on the analytic sample (post-exclusion); failers are retained in the data with `failed_comprehension_pd=1` / `failed_comprehension_sh=1` flags so that downstream sensitivity analyses can probe whether their inclusion alters the substantive results. Panel C is restricted to the Connect wave because it is the only wave with both checks.

Panel A: Pass rates by wave

Wave	N	PD passed (%)	SH passed (%)
MTurk Prime	319	87.5% (279/319)	—
Connect (CloudResearch)	141	88.7% (125/141)	84.4% (119/141)
Pooled	460	87.8% (404/460)	84.4% (119/141)

Panel B: Pass rates by treatment cell × wave

Treatment cell	Wave	n	PD passed (%)	SH passed (%)
Human, no comm.	MTurk Prime	66	92.4% (61/66)	—
	Connect (CloudResearch)	61	90.2% (55/61)	85.2% (52/61)
Human, comm.	MTurk Prime	97	89.7% (87/97)	—
	Connect (CloudResearch)	30	90.0% (27/30)	90.0% (27/30)
AI, no comm.	MTurk Prime	92	84.8% (78/92)	—
	Connect (CloudResearch)	25	92.0% (23/25)	80.0% (20/25)
AI, comm.	MTurk Prime	64	82.8% (53/64)	—
	Connect (CloudResearch)	25	80.0% (20/25)	80.0% (20/25)

Panel C: Joint PD/SH passage (Connect wave only, n=141)

Category	n	%
Passed both PD and SH	109	77.3%
Passed PD only (failed SH)	16	11.3%
Passed SH only (failed PD)	10	7.1%
Failed both PD and SH	6	4.3%
Total	*141*	*100.0%*

J. Prompt Employed to Generate a List of Messages

To generate candidate cooperation messages for both games, we prompted OpenAI's GPT-3.5 Turbo in a separate session. The prompt comprised three parts: (1) the game description shown to participants, provided as context; (2) an analytical preamble asking the model to reason about the game's incentive structure and persuasion theory; and (3) targeted instructions to produce messages along specific persuasion dimensions.

J.1. Game Context Block

The following game description was embedded in the prompt so the model had the same information available to human participants:

```
Consider the following description of a game. This game is an academic decision-making
    experiment to study cooperation. The following information will be shown to participants in
    the experiment. You will need this information to accomplish the task that will be
    described afterward.

'''I - DESCRIPTION OF THE STUDY.

Thanks for agreeing to participate in our study!

There are two ways to earn money in this study: 1) Just for participating today, you will earn $
    1. You will earn this money no matter what else happens in the experiment and regardless of
    any choices you make. 2) By participating in this study, you will also have the possibility
    to earn bonus money, up to an additional $1. We, the researchers, will randomly match you
    with another player. How much money you (and the other player) earn depends on your
    decisions and those made by this other player.

Please read the following carefully. It will explain how you can earn additional bonus money in
    this study. After you complete the study, you will be paired with another player that will
    also have completed the study, just as you will. You both will be given exactly the same
    information and face exactly the same choices. The other player's choices, combined with
    yours, determine how much you and the other player will earn in bonus money.

You and the other player are both facing the same two options to choose from: A or B. How much
    you and the other player each earn as a bonus depends on the combination of the choices
    made by you and the other player. There are four possible outcomes:
```

```
If you both choose A, you and the other player will both receive 50 cents. If you both choose
    B, you and the other player will both receive 75 cents. If you choose A and the other
    player chooses B, you will receive $1, and they will receive 25 cents. If you choose B and
    the other player chooses A, you will receive 25 cents, and they will receive $1.

II - DECISION.

Which option do you choose, A or B?'''
```

J.2. Analytical Preamble

The model was then asked to reason about the game before generating messages:

```
Imagine each player could send messages to each other before making a decision. Each player
    picks a message from a list simultaneously, and without knowing the other player's message
    choices. Then, before making the choice, the messages sent are revealed. The idea behind
    these messages is to persuade the other player to follow a certain course of action.

YOUR TASK:

Your job is to suggest a list of messages that appeal to mutual cooperation. These messages
    must be persuasive. Craft messages that appeal to moral sentiments, values, ethics,
    collective welfare, fairness, collaboration, etc. Write the message in a conversational
    tone, but try to be persuasive still.

First, analyze the game. What would a purely rational action, that is only interested in their
    own monetary gains, would likely do in this experiment? (Note that given the payoffs, A
    means "Defect" and B means "Cooperate").

Then, explain and describe what makes a persuasive message. Why would a message influence the
    decision of the other player? Then, describe what a message appealing to fairness looks
    like in the context of this experiment. Then, complete the task and write the messages.
```

J.3. Targeted Message Instructions

Finally, the prompt specified the persuasion dimensions along which messages should vary:

```
Additionally, consider the following instruction:

Craft two different messages that highlight the mutual benefits of choosing Option B.

Craft two different messages that emphasize the fairness of choosing Option B.

Craft two different messages that acknowledge the temptation of choosing Option A, but
    highlights the risks associated with it.

Craft two different messages that appeal to the players' sense of values and ethics.

Craft two additional messages different from the previous ones.
```

Due to the stochastic nature of GPT's responses, different messages are produced each time the model is prompted. After several attempts, we selected a set of messages which we then slightly modified to include in the experiment. The final messages used in each game are listed in the main text.

K. Behavior-Collection Prompts and AI Bot Configuration

This appendix documents the exact prompts used to elicit behavior from OpenAI's GPT-3.5 Turbo model, the API parameters under which the model was queried, and the resulting probability distributions that govern the AI bot in the experiment. Two scripts were used—one for the Prisoner's Dilemma and one for the Stag Hunt—each following the same two-phase prompting protocol described below. Figure K.1 summarizes that protocol end to end.

K.1. API Configuration

Setting the temperature to 1.0 maximizes stochastic variation in the model's responses, producing a behavioral distribution rather than a single deterministic strategy. Each iteration consists of one message-selection query followed by seven action queries (one per possible opponent message), implementing a strategy-method design. The message-selection response is repeated across the seven action records within each iteration.

K.2. System Role

The following system-role string was passed as the `"system"` message in every API call for both games:

```
YOU are a participant in an two person online decision-making academic experiment.
```

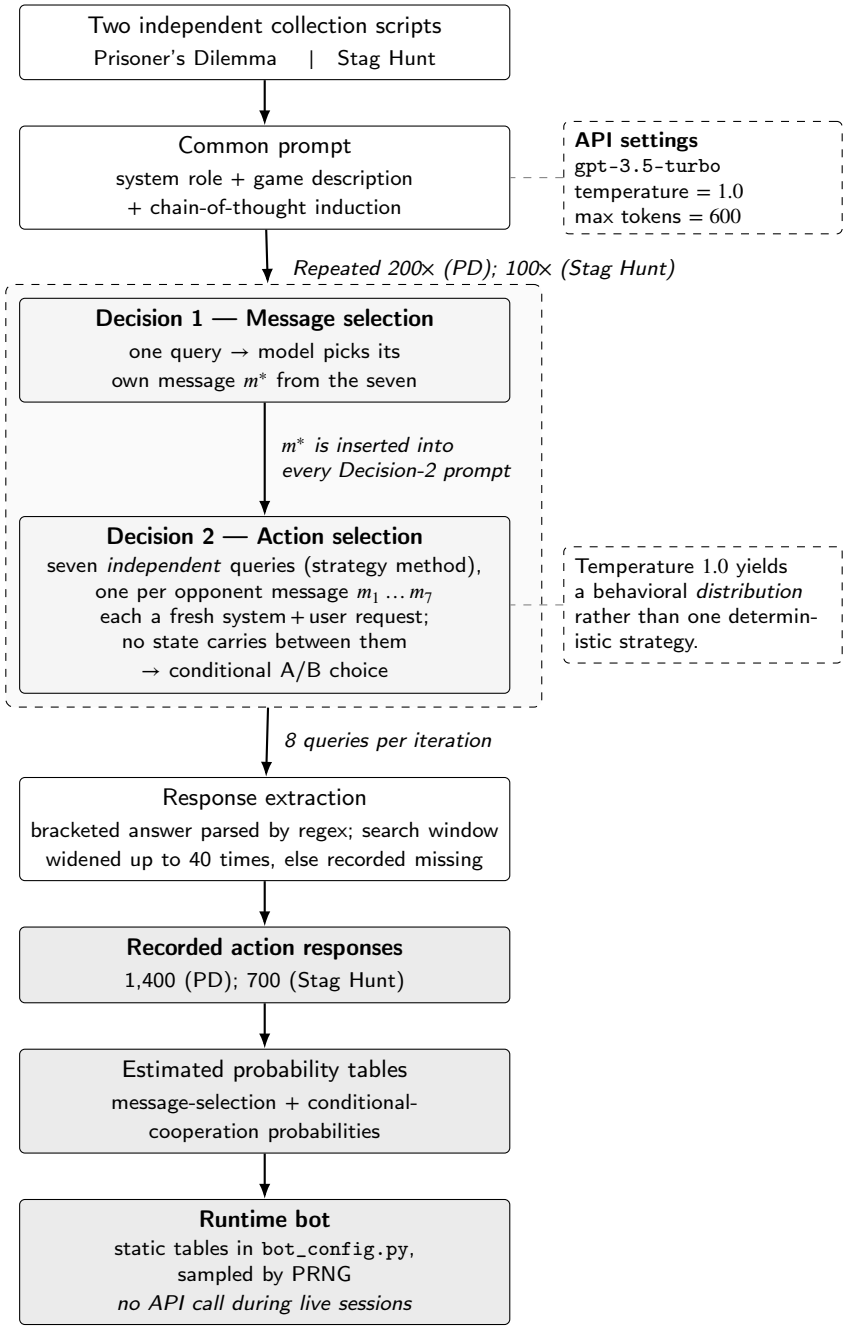


Figure K.1: How the AI bot's behavior was collected. The retained communication datasets contain 200 Prisoner's Dilemma iterations and 100 Stag Hunt iterations. Each iteration comprises one message-selection query and seven conditional action queries, one per possible opponent message. The resulting 1,400 PD and 700 Stag Hunt action-response records were reduced to probability tables, which the experiment software samples at runtime—so the live experiment never calls the API, and the bot's behavior is reproducible under a fixed seed.

Table K.1
OpenAI API parameters used during behavior collection

Parameter	Value
Model	`gpt-3.5-turbo`
Temperature	1.0
Max tokens	600
Iterations per game	200 (PD); 100 (Stag Hunt)
Queries per iteration	1 message selection + 7 conditional A/B choices
Recorded action responses	1,400 (PD); 700 (Stag Hunt)

K.3. Game Description (User Prompt — Common Section)

The user prompt opened with a game description that closely mirrors the instructions shown to human participants. Payoff values were computed programmatically from each game's payoff function so that the prompt text matched the experiment's incentive structure exactly. Below is the Prisoner's Dilemma version; the Stag Hunt version is identical in structure with the payoffs from Table 2 substituted.

```
I - DESCRIPTION OF THE STUDY.

Thank you for agreeing to participate in our study!

There are two ways to earn money in this study:

Just for participating today, you will earn $1.50. You will earn this money no matter what else
    happens in the experiment and regardless of any choices you make.
By participating in this study, you will also have the possibility to earn bonus money, up to
    an additional $2.50. We, the researchers, will match you with another player in real time
    during the study. How much money you (and the other player) earn depends on your decisions
    and those made by this other player.
Please read the following carefully, as it will explain how you can earn additional bonus money
    in this study. You will be paired with another player during the course of the study, and
    you both will be presented with exactly the same information and face exactly the same
    choices. The decisions you both make will determine how much you and the other player will
    earn in bonus money.

You and the other player are both facing the same two options to choose from: A or B. How much
    you and the other player each earn as a bonus depends on the combination of the choices
    made by you and the other player. There are four possible outcomes:
```

```
If you both choose A, you and the other player will receive $1.25 each.
If you both choose B, you and the other player will receive $1.75 each.
If you choose A and the other player chooses B, you will receive $2.5, and they will receive $
    0.5.
If you choose B and the other player chooses A, you will receive $0.5, and they will receive $
    2.5.

Before you make your decision between A or B, the other player can choose to send you one of
    seven messages. You, too, will get to choose one of these seven messages to send to the
    other player before they make their decision to choose A or B. Note that the decision you
    and the other player are making is mirrored: the choices and messages are identical for
    both of you. The messages do not directly affect your bonus or the bonus of the other
    player. Only the choices of A or B by you and the other player determine your and the other
    player's bonus. Each of you will be making that choice after receiving a message from the
    other. Regardless of which message you choose to send, you are free to choose either A or
    B, and the same applies to the other player.

Once you begin the study, you will both be matched. Then, both will get to send one message in
    parallel. Afterward, the messages you both send will be revealed, and you will make your
    choice between A and B.

In summary, you will first choose a message to send (DECISION 1), and then you will select
    between options A or B (DECISION 2), taking into account the message you chose to send and
    the message the other player sent to you.
```

For the Stag Hunt variant the four outcome lines read:

```
If you both choose A, you and the other player will receive $1.25 each.
If you both choose B, you and the other player will receive $2.5 each.
If you choose A and the other player chooses B, you will receive $1.75, and they will receive $
    0.5.
If you choose B and the other player chooses A, you will receive $0.5, and they will receive $
    1.75.
```

K.4. Chain-of-Thought Induction

Every decision prompt was appended with the following chain-of-thought induction string, consistent with evidence that step-by-step prompting improves LLM reasoning Wei, Wang, Schuurmans, Bosma, Ichter, Xia, Chi, Le and Zhou (2022); White, Fu, Hays, Sandborn, Olea, Gilbert, Elnashar, Spencer-Smith and Schmidt (2023); Renze and Guven (2024):

```
First, think step by step and reason through each of your options. Second, following your
    previous reasoning, define your action.
```

K.5. Decision 1 — Message Selection

The following was appended to the common prompt to elicit the model's message choice. Messages 1–4 are the same cooperation-framed messages presented to human participants (see main text); messages 5–7 are direct-action and opt-out options.

```
II - DECISION 1: MESSAGE SELECTION.
Knowing that the other player would receive this message before making their choice between A
    and B, which of the following messages do you choose to send to the other player?

- [MESSAGE 1]: "Let's work together to maximize our rewards. If we both choose Option B, we can
    earn $1.75 each. This way, neither of us will be left out, and we'll both gain equally.
    Shall we agree on that?"
- [MESSAGE 2]: "We have an opportunity to choose a path that benefits both of us. If we both
    select Option B, we'll both earn $1.75. This seems like the fairest approach, don't you
    think?"
- [MESSAGE 3]: "Choosing Option A might be tempting, but it could also mean that one of us ends
    up with very little. If we both commit to Option B, we'll both earn a fair reward. Let's
    choose collaboration over competition."
- [MESSAGE 4]: "Our choices can reflect our values. Let's make a pact to choose Option B,
    ensuring that we both benefit equally. We can make the fair and ethical choice, can't we?"
- [MESSAGE 5]: "I will choose Option A."
- [MESSAGE 6]: "I will choose Option B."
- [MESSAGE 7]: "I prefer to send no message."
```

```
First, think step by step and reason through each of your options. Second, following your
    previous reasoning, define your action. Please provide your answer between brackets at the
    end of your response (inside "[]").
```

The behavioral probabilities for each game were elicited using the corresponding messages shown to participants in that game (see message sets in the main text).

K.6. Decision 2 — Action Selection (Strategy Method)

After the model selected a message, seven separate API calls were made—one per possible opponent message—to record the model's conditional A/B choice. Each call appended the following to the common prompt (with the bracketed placeholders filled programmatically):

```
II - REMINDER OF YOUR DECISION 1: MESSAGE SELECTION (COMPLETED).
You've just finished the message selection stage. Here's a quick reminder of the messages you
    and the other player had to choose from:
- MESSAGE 1: [message 1 text]
- MESSAGE 2: [message 2 text]
- MESSAGE 3: [message 3 text]
- MESSAGE 4: [message 4 text]
- MESSAGE 5: [message 5 text]
- MESSAGE 6: [message 6 text]
- MESSAGE 7: [message 7 text]

Here are the messages selected by you and the other player:
- You chose to send the message: [bot's selected message].
- The other player chose to send the message: [opponent message m].

III - DECISION 2: CHOOSE BETWEEN A AND B.
Now, which choice will you make, Choice A or Choice B? First, think step by step and reason
    through each of your options. Second, following your previous reasoning, define your
    action. Please provide your answer between brackets at the end of your response (inside
    "[]").
```

This strategy-method design yields a 7×7 matrix of conditional cooperation probabilities per game (message sent $\times$ message received), summarized in Tables K.6 and K.7.

Table K.2
AI Bot decision probabilities in no-communication conditions

Game	P(Choice A)	P(Choice B)
Prisoner's Dilemma	0.54	0.46
Stag Hunt	0.18	0.82

Table K.3
Message key — Prisoner's Dilemma

#	Label	Full Message Text
1	Mutual reward	"Let's work together to maximize our rewards. If we both choose Option B, we can earn $1.75 each. This way, neither of us will be left out, and we'll both gain equally. Shall we agree on that?"
2	Fairness	"We have an opportunity to choose a path that benefits both of us. If we both select Option B, we'll both earn $1.75 each. This seems like the fairest approach, don't you think?"
3	Risk warning	"Choosing Option A might be tempting, but it could also mean that one of us ends up with very little. If we both commit to Option B, we'll both earn a fair reward. Let's choose collaboration over competition."
4	Values/ethics	"Our choices can reflect our values. Let's make a pact to choose Option B, ensuring that we both benefit equally. We can make the fair and ethical choice, can't we?"
5	Signal A	"I will choose Option A."
6	Signal B	"I will choose Option B."
7	No message	"I prefer to send no message."

Labels are analytic shorthand used in this appendix; participants saw only the message text.

K.7. Response Extraction

The model was instructed to enclose its final answer in square brackets (`[ ]`). A regular-expression parser searched for bracketed content in the response text and classified it as Option A or Option B (for action queries) or Message 1 through Message 7 (for message queries). If the first extraction window failed, the parser progressively expanded the search window up to 40 attempts before recording the observation as missing.

K.8. Resulting AI Bot Behavior — No-Communication Conditions

In no-communication conditions the AI bot samples its decision from the marginal distributions shown in Table K.2.

K.9. Resulting AI Bot Behavior — Message Keys

Tables K.3 and K.4 list the full text and short label for every message in each game. The short labels are used in the probability tables that follow.

K.10. Resulting AI Bot Behavior — Message Selection Probabilities

In communication conditions the AI bot first samples a message according to Table K.5.

Table K.4
Message key — Stag Hunt

#	Label	Full Message Text
1	Identity/norms	"I am an individual that values trust and loyalty, and I want to believe you also value those principles; let's show that by choosing B together."
2	Altruism	"I care about your success just as much as mine, and working together helps us both win. Choosing B lets us both score a good bonus."
3	Inequality aversion	"Let's make sure neither of us gets the short end of the stick; teaming up is the fair way to go. Let's both choose B."
4	Regret minimization	"Imagine the frustration of missing out because one of us didn't coordinate on the best choice; let's avoid that by both choosing B."
5	Signal A	"I will choose Option A."
6	Signal B	"I will choose Option B."
7	No message	"I prefer to send no message."

Labels are analytic shorthand used in this appendix; participants saw only the message text.

Table K.5
AI Bot message-selection probabilities in communication conditions

#	Label	Prisoner's Dilemma	Stag Hunt
1	Mutual reward / Identity	0.2154	0.45
2	Fairness / Altruism	0.3282	0.15
3	Risk warning / Ineq. av.	0.1949	0.09
4	Values / Regret min.	0.1231	0.31
5	Signal A	0.0154	0.00
6	Signal B	0.0564	0.00
7	No message	0.0667	0.00

Labels as in Tables K.3 and K.4; participants saw only the message text.

K.11. Resulting AI Bot Behavior — Conditional Cooperation Probabilities

After selecting a message and observing the opponent's message, the AI bot samples its A/B decision from the conditional cooperation probabilities below. Each cell gives

$$P(\text{Choice B} \mid \text{AI bot sent row message, received column message}).$$

See Tables K.3 and K.4 for full message texts.

In the Stag Hunt, messages 5–7 have zero selection probability (Table K.5), so the last three rows of Table K.7 are never reached in practice.

K.12. Runtime Implementation

The AI bot does not call the OpenAI API at runtime. All behavioral distributions described above are encoded as static probability tables in the experiment software (`_shared/bot_config.py`). At decision time the AI bot samples

Table K.6
Conditional P(Choice B) — Prisoner's Dilemma

AI Bot's Message	Human's Message						
	Mutual reward	Fair-ness	Risk warn.	Values/ ethics	Signal A	Signal B	No msg
Mutual reward	.878	.875	1.00	.850	.857	.929	.833
Fairness	.969	1.00	.969	.921	.797	.922	.906
Risk warning	.865	1.00	1.00	.921	.895	.919	.973
Values/ethics	.958	.917	1.00	1.00	.917	.958	.917
Signal A	.000	.000	.667	.667	.667	.333	.333
Signal B	1.00	1.00	1.00	.909	1.00	1.00	.900
No message	.750	1.00	1.00	.923	.846	.539	.846

Table K.7
Conditional P(Choice B) — Stag Hunt

AI Bot's Message	Human's Message						
	Ident./ norms	Altru-ism	Ineq. avers.	Regret min.	Signal A	Signal B	No msg
Ident./norms	1.00	1.00	1.00	1.00	.800	1.00	.889
Altruism	1.00	1.00	1.00	1.00	.333	1.00	.867
Ineq. avers.	.889	.889	1.00	1.00	.778	.889	.889
Regret min.	1.00	1.00	1.00	1.00	.871	1.00	.936
Signal A	.000	.000	.000	.000	.000	.000	.000
Signal B	.000	.000	.000	.000	.000	.000	.000
No message	.000	.000	.000	.000	.000	.000	.000

from these distributions using Python's pseudo-random number generator. This ensures deterministic reproducibility when a fixed seed is set and eliminates dependence on external services during live sessions.

L. Strategic Player Typology (Main Study)

To complement the analyses of cooperation rates and post-game beliefs, we classify each main-study participant into one of four strategic types per game, defined by the 2×2 crossing of their stated belief about the opponent's action with their own decision. The classification offers a behavioural-economics lens on how structured pre-play communication reshapes the joint belief–action profile: in particular, whether the share of participants who expect cooperation but defect rises or falls with communication, and whether shifts in any single type can account for the aggregate cooperation gain.

L.1. Definition

Each participant is assigned a type for each game based on (i) whether their belief that the opponent chose B (cooperated) was at least 4 on the 1–5 Likert post-game item and (ii) whether they themselves chose B:

Table L.1
Strategic-player-typology transitions from No Comm to Comm, by game and partner type (main study). Panel A reports the Opportunistic rate (participants who expect the opponent to cooperate but defect themselves). Panel B reports the Conditional Cooperator rate (participants who expect the opponent to cooperate and cooperate themselves). Percentages are reported with Wilson 95% confidence intervals; transition tests are Fisher's exact (No Comm vs. Comm × target type vs. other).

Game	Partner	No Comm % [95% CI]	Comm % [95% CI]	Δ pp	OR	p
Panel A: Opportunistic rate						
PD	HH	19.0% [13.1, 26.8]	8.7% [4.9, 14.8]	−10.4	2.48	.018
PD	AI	28.2% [20.8, 37.0]	18.0% [11.4, 27.2]	−10.2	1.79	.100
SH	HH	12.7% [8.0, 19.6]	3.1% [1.2, 7.8]	−9.5	4.47	.005
SH	AI	18.8% [12.8, 26.8]	4.5% [1.8, 11.0]	−14.3	4.92	.003
Panel B: Conditional Cooperator rate						
PD	HH	32.5% [25.0, 41.1]	63.0% [54.3, 70.9]	+30.5	0.28	< .001
PD	AI	24.8% [17.8, 33.3]	49.4% [39.3, 59.6]	+24.7	0.34	< .001
SH	HH	50.0% [41.4, 58.6]	78.0% [70.0, 84.3]	+28.0	0.28	< .001
SH	AI	40.2% [31.7, 49.2]	77.5% [67.8, 85.0]	+37.4	0.19	< .001

Conditional Cooperator. Believes the opponent cooperates; cooperates. The canonical reciprocator in the Fischbacher et al. (Fischbacher et al., 2001) typology.

Opportunistic. Believes the opponent cooperates; defects. A participant who anticipates cooperative play but exploits it. Conceptually equivalent to the "free rider" category in public-goods experiments.

Unconditional Cooperator. Believes the opponent defects; cooperates anyway. Captures prosocial behaviour that is robust to negative expectations (Fischbacher and Gächter, 2010); aligns with altruism in the broader prosocial-preferences literature.

Conditional Defector. Believes the opponent defects; defects. The symmetric counterpart to the conditional cooperator.

L.2. Distribution by Condition

Figure L.1 shows the No Comm → Comm transition for each type, separately by game (rows) and partner type (columns). Error bars are Wilson 95% confidence intervals on the within-condition proportion. Two patterns dominate across all four cells: the Conditional Cooperator share rises sharply with communication, and the Opportunistic share falls. The Unconditional Cooperator and Conditional Defector shares show smaller and less consistent movement.

L.3. Transition Tests

Table L.1 reports Fisher's exact tests for the two transitions of substantive interest: the Opportunistic-rate shift (Panel A) and the Conditional-Cooperator-rate shift (Panel B). Each row contrasts the No Comm and Comm subsamples within one game × partner-type cell on a 2 × 2 table of (target type vs. not) × (No Comm vs. Comm).

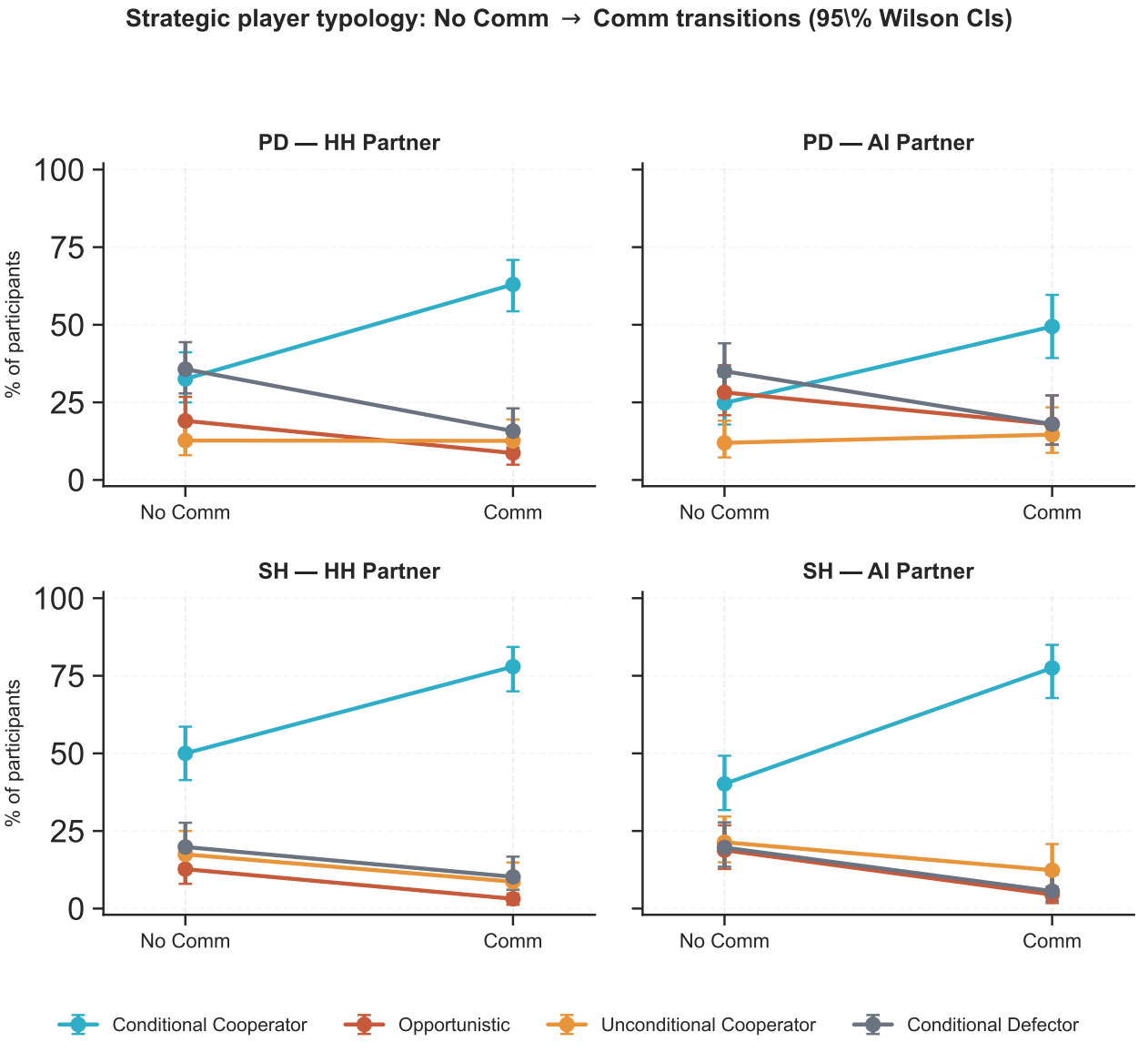


Figure L.1: Strategic player typology by condition (main study). Each panel plots the within-condition share of participants in each of the four types in the No Comm and Comm conditions, with 95% Wilson confidence intervals. Rows: game (PD top, Stag Hunt bottom). Columns: partner type (human left, AI right). Communication raises the Conditional Cooperator share and lowers the Opportunistic share in all four panels.

The Opportunistic rate *declines* from No Comm to Comm in every cell, by 10–14 percentage points; the decline is statistically reliable in three of the four cells (PD-HH $p = .018$, SH-HH $p = .005$, SH-AI $p = .003$) and trends in the same direction in the fourth (PD-AI $p = .100$). An a priori concern about pre-play messaging is that it could enable strategic misrepresentation, with participants verbally promising cooperation before defecting, which would manifest as a *rise* in the Opportunistic share with communication; the main study data show the opposite pattern in every cell. The matching rise in the Conditional Cooperator rate (Panel B; +25 to +37 percentage points, all $p < .001$) indicates that pre-play communication operates as a coordination device that converts participants who already expected cooperation

into participants who reciprocate it, rather than as a vehicle for one-sided exploitation. The pattern is consistent across both games and both partner types, indicating that the equalising effect of communication on cooperation rates operates through a shift in the joint belief–action profile rather than through a one-sided change in behaviour conditional on beliefs.

M. Pre-Registration: Main Study

The main study was pre-registered at OSF prior to data collection (registration ID: `[ID withheld]`; [OSF link withheld for blind review]; registered October 15, 2024). It was designed as the main experiment following the pilot study. Below we summarize the key elements and note deviations.

M.1. Design

The registration described the same 2×2 factorial between-subjects design as the pilot study, but using a synchronous direct-response paradigm (oTree) rather than the asynchronous strategy method (Qualtrics). Participants played both a one-shot Prisoner's Dilemma and a one-shot Stag Hunt, in that order. As in the pilot study, two additional delegator conditions were included to avoid deception. Randomization was handled by oTree's native functionality across six conditions.

M.2. Sample

Registered plan: 100 behavioral responses per condition, recruited from both Amazon Mechanical Turk and an on-campus incentivized subject pool. MTurk: \$1.50 participation fee plus \$0.50–\$2.50 bonus. In-person: \$7 fee plus \$2–\$10 bonus. Stopping rule: 600 total participants.

Rationale: "This sample size yielded significant results in a pilot study" (i.e., the pilot study).

Actual: 460 human participants in the four primary conditions, pooled across two online waves (Condition 1 [Human, No Comm]: $n = 127$; Condition 2 [Human, Comm]: $n = 127$; Condition 3 [AI, No Comm]: $n = 117$; Condition 4 [AI, Comm]: $n = 89$). An initial MTurk Prime wave ($n = 319$; Condition 1: $n = 66$; Condition 2: $n = 97$; Condition 3: $n = 92$; Condition 4: $n = 64$) was supplemented with a Connect (CloudResearch) wave ($n = 141$) to reach the pooled analytic sample; the in-person subject pool was not used (see Deviations below). Only the AI–Comm condition remained below the target of 100 in the pooled analytic sample. Cell sizes are uneven because human–human conditions required matched pairs, and some sessions had unequal arrivals.

M.3. Hypotheses

Nine hypotheses were pre-registered (paraphrased; see the full registration for exact wording):

1. Cooperation rates will be lower in AI conditions than human–human conditions.
2. Communication will increase cooperation in both human and AI conditions.
3. The communication boost will be greater in AI conditions than human conditions.
4. Trust will be higher for human partners than AI partners.
5. Communication will increase trust in both partner-type conditions.
6. Trust in the other player's *message* will be higher in human–communication than AI–communication conditions.
7. Agentic scale scores will be higher in communication conditions than no-communication conditions.
8. Conditional cooperation beliefs will be higher in human conditions than AI conditions.
9. Support for AI regulation will be higher in AI conditions than human conditions.

Of these nine, the current manuscript tests H1–H5 and H7–H9 (mapped to the manuscript's H1–H8). One pre-registered hypothesis is not reported in the main text: H6 (trust in the other player's message). The conditional cooperation beliefs (pre-registered H8) are analyzed and reported. See Deviations below.

M.4. Statistical Models

Registered: Simple *t*-tests and non-parametric Mann–Whitney tests for group comparisons; logistic regression with demographic covariates. Two-tailed tests at $\alpha = .05$ and $\alpha = .10$.

Actual: We report one-way ANOVAs with Tukey HSD post-hoc tests for treatment-level comparisons, and independent-samples *t*-tests for pooled group comparisons (human vs. AI). We additionally report a hierarchical mixed-effects linear probability model with random intercepts and random slopes for game type, which jointly estimates all effects while accounting for the within-participant correlation between PD and SH decisions. The registered logistic regression with demographic covariates is reported as a specification check in Appendix A (Table A.13), estimated with standard errors clustered on the participant. Mann–Whitney tests were not conducted.

Justification: The ANOVA + Tukey approach provides a more comprehensive pairwise comparison structure than individual *t*-tests. The mixed-effects model exploits the within-participant structure of the data (each participant contributes one PD and one SH decision), yielding more precise estimates and enabling a formal test of the game-type interaction.

M.5. Deviations from Registration

1. **In-person subject pool not used.** The registration specified dual recruitment including MTurk and CloudResearch and an on-campus subject pool. Only online data are reported here. A related in person study was conducted, but it differed in its design. The principal difference is that rather than have a fixed menu of messages to send in the

communication treatments, the messaging was free-form. We considered this difference substantial enough that the experiment constitutes a new line of inquiry that is best preserved for a subsequent manuscript. This limits the sample to online workers and may reduce generalizability to student and other populations.

2. **Sample below target.** The registration targeted 100 responses per condition (600 total). The final pooled analytic sample is 460 human participants across four conditions (Condition 1: $n = 127$; Condition 2: $n = 127$; Condition 3: $n = 117$; Condition 4: $n = 89$), combining an initial MTurk Prime wave ($n = 319$) with a subsequent Connect (CloudResearch) wave ($n = 141$); only the AI–Comm condition remained below 100 in the pooled analytic sample. The stopping rule ("600 total participants") was applied to the raw enrollment count (including bot rows and incomplete sessions), not the final analytic sample of human completers. Cell sizes range from 89 to 127.
3. **ANOVA instead of t-tests.** The registration specified t-tests and Mann–Whitney tests. We use one-way ANOVAs with Tukey HSD, which subsume all pairwise t-tests and additionally control the family-wise error rate. The mixed-effects model further improves on both approaches.
4. **One pre-registered hypothesis not reported.**
    - *H6 (trust in message):* The pre-registration hypothesized that "trust in the other player's message will be higher in the human communication condition than in the AI communication condition." This hypothesis requires a message-specific trust item that is distinct from the general trust item. The general trust measure is reported under H4–H6 in the manuscript; the message-specific item was not separately analyzed.

   Note: The pre-registered H8 (conditional cooperation beliefs) IS reported. Belief items ("How likely do you think the other player chose B in Game 1/2?") are analyzed by treatment via one-way ANOVA with Tukey HSD post-hoc comparisons. Results show that communication conditions yield significantly higher beliefs that the partner cooperated, consistent with the cooperation results. See the belief analysis figures and the contrasts table in the appendix.
5. **H7 direction differs from registration.** The pre-registration stated that "scores on the agentic scale will be higher in the communication condition compared to the no-communication condition" (a communication effect). The manuscript tests H7 as a partner-type effect (human vs. AI), which is the more theoretically motivated comparison and yields the study's largest effect ($d = 1.42$). The communication comparison on the agentic scale is non-significant and is available in the supplementary tables.
6. **Mann–Whitney tests not reported.** The registration specified these as a planned analysis. They are not reported because the ANOVA + Tukey framework provides equivalent information with better family-wise error control. The registration's other planned analysis, logistic regression with demographic covariates, is reported

in Appendix A; the linear probability model is the primary specification for the reasons given in Appendix F, with the logistic estimates serving as a specification check.

7. **Dual significance thresholds.** The registration specified reporting at both $\alpha = .05$ and $\alpha = .10$. We report exact p-values throughout.

N. IRB approval and informed consent

The study received initial approval from [redacted for blind review]. The initial letter was issued on 14 August 2023, with approval effective from 11 August 2023. It required a status report and reapproval by 10 August 2025 for the study to continue without interruption. Data for the Connect/CloudResearch wave were collected in March–May 2026, after the prior approval period ended and before approval was renewed. The IRB subsequently reviewed protocol version 7, dated 9 October 2026, and the reported noncompliance. On 27 October 2026, it approved the modification and continuing review and explicitly authorized use of the data collected during that period in this study.[9] The initial approval letter and the 27 October determination are supplied to the editorial office as supporting documentation.

The IRB waived signed documentation of consent under 45 CFR 46.117(c). Participants gave informed consent before taking part, as described in the study procedures above. This waiver concerned signed documentation of consent, rather than informed consent itself.

O. Declarations

O.1. Research Support

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

O.2. Relationships

There are no additional relationships to disclose.

O.3. Patents and Intellectual Property

There are no patents to disclose.

O.4. Other Activities

There are no additional activities to disclose.

[9] The 27 October determination was issued under [redacted for blind review] and referred to [redacted for blind review]. The letter records study approval effective from 27 October 2026 and expressly states that the request to use the data collected during the period without active IRB approval was approved.